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United States Patent	12404558
Kind Code	B2
Date of Patent	September 02, 2025
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Methods and assays for detection and subtyping of microbial pathogens

Abstract

The present invention provides methods of detecting a biothreat agent in a sample, comprising detecting at least one biothreat-specific amplicon in the sample. The methods also encompass confirming the absence of the biothreat agent by detecting Near Neighbor specific amplicons to avoid false positive results.

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Appl. No.:	17/426600
Filed (or PCT Filed):	January 28, 2020
PCT No.:	PCT/US2020/015395
PCT Pub. No.:	WO2020/159977
PCT Pub. Date:	August 06, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220090172 A1	Mar. 24, 2022

Related U.S. Application Data

us-provisional-application US 62798463 20190129

Publication Classification

Int. Cl.: **C12Q1/689** (20180101); **C12Q1/04** (20060101); **C12Q1/6844** (20180101); **C12Q1/686** (20180101); **C12R1/07** (20060101)

U.S. Cl.:

CPC **C12Q1/689** (20130101); **C12Q1/04** (20130101); **C12Q1/6844** (20130101); **C12Q1/686** (20130101); C12Q2600/156 (20130101); C12Q2600/16 (20130101); C12R2001/07 (20210501)

Field of Classification Search

CPC: C12Q (1/689)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is the U.S. National Stage of International Application No. PCT/US2020/015395, filed on Jan. 28, 2020, which claims priority to U.S. Provisional Patent Application No. 62/798,463, filed on Jan. 29, 2019, the contents of each of which are incorporated herein by reference in their entireties.

REFERENCE TO SEQUENCE LISTING SUBMITTED ELECTRONICALLY

(1) The official copy of the sequence listing is submitted electronically via EFS-Web as an ASCII-formatted sequence listing with a file named “91482_239PCT_SeqList_ST25.txt” created on Jan. 27, 2020 and having a size of 83.2 kilobyte, and is filed concurrently with the specification. The sequence listing contained in this ASCII-formatted document is part of the specification and is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(2) This invention relates to methods, primers, assays, and kits for detecting the presence of microbial pathogens in a sample.

BACKGROUND

(3) Throughout recent history, various aggressor nation states and terrorist groups have shown the willingness and/or capability to develop and use biological weapons against war fighters and civilian populations. The ability to detect the agents being developed as well as their virulence and antibiotic resistance profiles, in environmental and clinical materials, would further our capability to detect the development of these agents and their use.

(4) The goal of several federal biosurveillance projects has been the early detection of biothreat agents to prevent or curtail mass civilian or military casualties. These systems have relied upon real-time-PCR to give a binary answer of presence or absence of the target. One challenge has been the complexity of the environmental samples, where tens of thousands of microorganisms exists, many of which are highly similar to the target pathogens. BioWatch is an example where numerous false positive results have been generated due to poorly known near-neighbor species confusing individual assays. While our knowledge of near-neighbors and of the target Biothreat agents is rapidly increasing, it is unrealistic to ever expect complete knowledge of either. DNA sequencing offers great potential, and there is a need for primers, methods, assays, and kits with greater ability to discriminate microbial pathogens in complex environmental and clinical sample matrices.

SUMMARY

(5) Timely and accurate detection and characterization of bacterial biothreat agents is vital for our nation's safety. Current systems for early detection of these agents rely upon single locus Polymerase Chain Reaction (PCR) methods, giving only presence/absence results. This methodology can and has led to false positives due to limited signature validation. The Inventors have developed a multi-agent multi-locus amplicon sequencing protocol encompassing 79 targets aimed at detecting the presence or absence of 5 biothreat agents, as well as the presence and sequence of plasmids, virulence factors, antimicrobial resistance factors, and sequence variant loci for Near Neighbor species differentiation. The agents targeted are *Burkholderia pseudomallei*, *Burkholderia mallei*, *Bacillus anthracis*, *Yersinia pestis*, and *Francisella tularensis*.

(6) The multi-agent assay, consisting of two multiplex amplification reactions, was validated against a diverse subset of target agent and near neighbor panels that were previously used to validate assays targeting individual agents. These panels consisted of 10-14 target agent strains and 11-48 NN strains. Sensitivity was 100% for all target agents, specificity was 91-100%. Targeted amplicon sequencing utilizing a universal amplicon indexing scheme provides a superior alternative to the current single locus PCR systems and enables the detection of multiple biothreat agents across multiple samples with a single sequencing run.

(7) In certain aspects, the present invention provides A method of detecting *Bacillus anthracis* in a sample, comprising detecting at least one *B. anthracis*-specific amplicon in the sample using at least one primer pair selected from the group consisting of: SEQ ID NO: 1 and SEQ ID NO: 2; SEQ ID NO: 3 and SEQ ID NO: 4; SEQ ID NO: 5 and SEQ ID NO: 6; SEQ ID NO: 7 and SEQ ID NO: 8; SEQ ID NO: 9 and SEQ ID NO: 10; SEQ ID NO: 11 and SEQ ID NO: 12; SEQ ID NO: 13 and SEQ ID NO: 14; SEQ ID NO: 15 and SEQ ID NO: 16; SEQ ID NO: 17 and SEQ ID NO: 18; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the *B. anthracis*-specific amplicon indicates the presence of *B. anthracis* in the sample, and the absence of the *B. anthracis*-specific amplicon indicates the absence of *B. anthracis* from the sample.

(8) In other aspects, the method further comprises confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 19 and SEQ ID NO: 20; SEQ ID NO: 21 and SEQ ID NO: 22; SEQ ID NO: 23 and SEQ ID NO: 24; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. anthracis* Near Neighbor-specific

amplicon in the sample confirms the absence of *B. anthracis*.

(9) In yet other aspects, the method further comprises confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific sequence variant (SV) or single nucleotide polymorphism (SNP) using at least one primer pair selected from the group consisting of: SEQ ID NO: 25 and SEQ ID NO: 26; SEQ ID NO: 27 and SEQ ID NO: 28; SEQ ID NO: 49 and SEQ ID NO: 50; SEQ ID NO: 51 and SEQ ID NO: 52; SEQ ID NO: 53 and SEQ ID NO: 54; SEQ ID NO: 55 and SEQ ID NO: 56; SEQ ID NO: 57 and SEQ ID NO: 58; SEQ ID NO: 59 and SEQ ID NO: 60; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. anthracis* Near Neighbor-specific SV in the sample confirms the absence of *B. anthracis*.

(10) In some aspects, the method further comprises detecting a virulence locus or virulence plasmid in the sample by detecting a virulence-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 29 and SEQ ID NO: 30; SEQ ID NO: 31 and SEQ ID NO: 32; SEQ ID NO: 33 and SEQ ID NO: 34; SEQ ID NO: 35 and SEQ ID NO: 36; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the virulence-specific amplicon indicates the presence of the virulence locus or virulence plasmid in the sample.

(11) In other aspects, the method further comprises detecting at least one drug resistance single nucleotide polymorphism (SNP) from *B. anthracis* in the sample using at least one primer pair selected from the group consisting of: SEQ ID NO: 37 and SEQ ID NO: 38; SEQ ID NO: 39 and SEQ ID NO: 40; SEQ ID NO: 41 and SEQ ID NO: 42; SEQ ID NO: 43 and SEQ ID NO: 44; SEQ ID NO: 45 and SEQ ID NO: 46; SEQ ID NO: 47 and SEQ ID NO: 48; a pair of sequences which are at least 85% identical thereto; and RNA equivalents. In other aspects, the method further comprises detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in the sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon uses at least one primer pair selected from the group consisting of: SEQ ID NO: 61 and SEQ ID NO: 62; SEQ ID NO: 63 and SEQ ID NO: 64; SEQ ID NO: 65 and SEQ ID NO: 66; SEQ ID NO: 67 and SEQ ID NO: 68; SEQ ID NO: 69 and SEQ ID NO: 70; SEQ ID NO: 71 and SEQ ID NO: 72; SEQ ID NO: 73 and SEQ ID NO: 74; SEQ ID NO: 75 and SEQ ID NO: 76; SEQ ID NO: 77 and SEQ ID NO: 78; SEQ ID NO: 79 and SEQ ID NO: 80; SEQ ID NO: 81 and SEQ ID NO: 82; SEQ ID NO: 83 and SEQ ID NO: 84; SEQ ID NO: 85 and SEQ ID NO: 86; SEQ ID NO: 87 and SEQ ID NO: 88; SEQ ID NO: 89 and SEQ ID NO: 90; SEQ ID NO: 91 and SEQ ID NO: 92; SEQ ID NO: 93 and SEQ ID NO: 94; SEQ ID NO: 95 and SEQ ID NO: 96; SEQ ID NO: 97 and SEQ ID NO: 98; SEQ ID NO: 99 and SEQ ID NO: 100; SEQ ID NO: 101 and SEQ ID NO: 102; SEQ ID NO: 103 and SEQ ID NO: 104; SEQ ID NO: 103 and SEQ ID NO: 104; SEQ ID NO: 105 and SEQ ID NO: 106; SEQ ID NO: 107 and SEQ ID NO: 108; SEQ ID NO: 117 and SEQ ID NO: 118; SEQ ID NO: 119 and SEQ ID NO: 120; SEQ ID NO: 121 and SEQ ID NO: 122; SEQ ID NO: 123 and SEQ ID NO: 124; SEQ ID NO: 125 and SEQ ID NO: 126; a pair of sequences which are at least 85% identical thereto; and RNA equivalents wherein the presence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

(12) In certain aspects, the method further comprises confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 177 and SEQ ID NO: 178; SEQ ID NO: 179 and SEQ ID NO: 180; SEQ ID NO: 181 and SEQ ID NO: 182; SEQ ID NO: 183 and SEQ ID NO: 184; SEQ ID NO: 185 and SEQ ID NO: 186; SEQ ID NO: 187 and SEQ ID NO: 188; SEQ ID NO: 189 and SEQ ID NO: 190; SEQ ID NO: 191 and SEQ ID NO: 192; SEQ ID NO: 193 and SEQ ID NO: 194; SEQ ID NO: 195 and SEQ ID NO: 196; SEQ ID NO: 197 and SEQ ID NO: 198; SEQ ID NO: 199 and SEQ ID NO: 200; SEQ ID NO: 201 and SEQ ID

NO: 202; SEQ ID NO: 203 and SEQ ID NO: 204; SEQ ID NO: 205 and SEQ ID NO: 206; SEQ ID NO: 207 and SEQ ID NO: 208; SEQ ID NO: 207 and SEQ ID NO: 208; SEQ ID NO: 209 and SEQ ID NO: 210; SEQ ID NO: 211 and SEQ ID NO: 212; SEQ ID NO: 213 and SEQ ID NO: 214; SEQ ID NO: 215 and SEQ ID NO: 216; SEQ ID NO: 217 and SEQ ID NO: 218; SEQ ID NO: 219 and SEQ ID NO: 220; SEQ ID NO: 221 and SEQ ID NO: 222; SEQ ID NO: 223 and SEQ ID NO: 224; SEQ ID NO: 225 and SEQ ID NO: 226; SEQ ID NO: 227 and SEQ ID NO: 228; SEQ ID NO: 229 and SEQ ID NO: 230; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

(13) In yet other aspects, the method further comprises confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV using at least one primer pair selected from the group consisting of: SEQ ID NO: 109 and SEQ ID NO: 110; SEQ ID NO: 111 and SEQ ID NO: 112; SEQ ID NO: 113 and SEQ ID NO: 114; SEQ ID NO: 115 and SEQ ID NO: 116; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

(14) In some aspects, the method further comprises detecting at least one drug resistance SNP or SV from *Burkholderia* spp. in the sample using at least one primer pair selected from the group consisting of: SEQ ID NO: 127 and SEQ ID NO: 128; SEQ ID NO: 129 and SEQ ID NO: 130; SEQ ID NO: 131 and SEQ ID NO: 132; SEQ ID NO: 133 and SEQ ID NO: 134; SEQ ID NO: 135 and SEQ ID NO: 136; SEQ ID NO: 137 and SEQ ID NO: 138; SEQ ID NO: 145 and SEQ ID NO: 146; SEQ ID NO: 147 and SEQ ID NO: 148; SEQ ID NO: 149 and SEQ ID NO: 150; SEQ ID NO: 151 and SEQ ID NO: 152; SEQ ID NO: 153 and SEQ ID NO: 154; a pair of sequences which are at least 85% identical thereto; and RNA equivalents.

(15) In other aspects, the method further comprises detecting *Francisella tularensis* in the sample by detecting at least one *F. tularensis*-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 265 and SEQ ID NO: 266; SEQ ID NO: 267 and SEQ ID NO: 268; SEQ ID NO: 269 and SEQ ID NO: 270; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is present in the sample, and an absence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is absent in the sample.

(16) In yet other aspects, the method further comprises confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 285 and SEQ ID NO: 286; SEQ ID NO: 287 and SEQ ID NO: 288; SEQ ID NO: 289 and SEQ ID NO: 290; SEQ ID NO: 291 and SEQ ID NO: 292; SEQ ID NO: 293 and SEQ ID NO: 294; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *F. tularensis* Near Neighbor-specific amplicon in the sample confirms the absence of *F. tularensis*.

(17) In one aspect, the method further comprises confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV using at least one primer pair selected from the group consisting of: SEQ ID NO: 271 and SEQ ID NO: 272; SEQ ID NO: 273 and SEQ ID NO: 274; SEQ ID NO: 275 and SEQ ID NO: 276; SEQ ID NO: 277 and SEQ ID NO: 278; SEQ ID NO: 279 and SEQ ID NO: 280; SEQ ID NO: 281 and SEQ ID NO: 282; SEQ ID NO: 283 and SEQ ID NO: 284; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *F. tularensis* Near Neighbor-specific SNP or SV in the sample confirms the absence of *F. tularensis*.

(18) In another aspect, the method further comprises detecting *Yersinia pestis* in the sample by detecting at least one *Y. pestis*-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 231 and SEQ ID NO: 232; SEQ ID NO: 233 and SEQ ID NO: 234; SEQ ID NO: 235 and SEQ ID NO: 236; SEQ ID NO: 237 and SEQ ID NO: 238; a pair of

sequences which are at least 85% identical thereto; and RNA equivalents; wherein the presence of the *Y. pestis*-specific amplicon indicates the presence of *Y. pestis* in the sample, and an absence of the *Y. pestis*-specific amplicon indicates an absence of *Y. pestis* in the sample.

(19) In still another aspect, the method further comprises confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV using at least one primer pair selected from the group consisting of: SEQ ID NO: 249 and SEQ ID NO: 250; SEQ ID NO: 251 and SEQ ID NO: 252; SEQ ID NO: 253 and SEQ ID NO: 254; SEQ ID NO: 255 and SEQ ID NO: 256; SEQ ID NO: 257 and SEQ ID NO: 258; SEQ ID NO: 259 and SEQ ID NO: 260; SEQ ID NO: 261 and SEQ ID NO: 262; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *Y. pestis* Near Neighbor-specific SNP or SV confirms the absence of *Y. pestis*.

(20) In certain aspects, the method further comprises confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 263 and SEQ ID NO: 264; a pair of sequences which are at least 85% identical thereto; and RNA equivalents; wherein detecting the *Y. pestis* Near Neighbor-specific amplicon confirms the absence of *Y. pestis*.

(21) In other aspects, the method further comprises characterizing and/or subtyping *Y. pestis* in the sample by detecting at least one amplicon using at least one primer pair selected from the group consisting of: SEQ ID NO: 239 and SEQ ID NO: 240; SEQ ID NO: 241 and SEQ ID NO: 242; SEQ ID NO: 243 and SEQ ID NO: 244; SEQ ID NO: 245 and SEQ ID NO: 246; SEQ ID NO: 247 and SEQ ID NO: 248; a pair of sequences which are at least 85% identical thereto; and RNA equivalents.

(22) In some aspects, the amplicons are generated with at least one multiplex amplification reaction. In other aspects, the amplicons are generated with at least two, at least three, at least four, or at least five multiplex amplification reactions.

(23) In other aspects, the amplicon, SNP or SV is determined using next-generation sequencing. In one aspect, each primer in the at least one primer pair comprises a universal tail sequence. In some aspects, the universal tail sequence comprises SEQ ID NO: 301 or SEQ ID NO: 303.

(24) In certain aspects, the amplicon is present when a locus read count of the amplicon is at least 10 sequence reads covering at least 75% of a corresponding amplicon reference sequence.

(25) In other aspects, sequence analysis of sequence read alignments is performed to determine whether a target species, Near Neighbor species, virulence or antibiotic resistance allele is present in the sample, wherein the target species is *Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, or *Yersinia pestis*.

(26) In one embodiment, the sample is an environmental sample. In another embodiment, the sample is a biological sample obtained from a subject.

(27) In certain embodiments, the method further comprises administering an effective amount of at least one antibiotic to the subject, wherein the at least one antibiotic is selected from the group consisting of a fluoroquinolone, an aminoglycoside, a glycopeptide, a lincosamide, a macrolide/ketolide, a cephalosporin, a monobactam, a nitroimidazole, a penicillin, a streptogramin, a tetracycline, and a physiologically acceptable salt, prodrug, or combination thereof.

(28) In another embodiment, the at least one antibiotic is not a fluoroquinolone if a *gyrA* drug resistance SNP is detected; and/or the at least one antibiotic is not a fluoroquinolone if a *parC* drug resistance SNP is detected; and/or the at least one antibiotic is not a fluoroquinolone or an aminocoumarin if a *gyrB* drug resistance SNP is detected; and/or the at least one antibiotic is not a rifamycin if a *rpoB* drug resistance SNP is detected; and/or the at least one antibiotic is not a β -lactam if a *penA* drug resistance SNP is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *folM* drug resistance SV is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeT* drug resistance SV is detected; and/or the at least one antibiotic is not a

trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a bpeS drug resistance SV is detected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B depict a universal index multiplex amplicon sequencing assay (UI-AmpSeq). Gene-specific primers containing universal tails (UT) amplify gene-specific targets. An ILLUMINA® extension PCR is run on these tailed PCR amplicons using ILLUMINA® indices containing complementary sequences to the UT, which allow for amplicon sequencing on an ILLUMINA® platform. FIG. 1A depicts sequence ready multi-locus amplification. The universal indexing strategy comprising universal tails is described in U.S. Publication No. 2016/0326572, the contents of which are incorporated herein by reference. FIG. 1B depicts sequencing and bioinformatic analysis.
- (2) FIG. 2 depicts an ASAP analysis. Enhanced Amplicon Sequencing Analysis pipeline (ASAP) allows the user to quickly analyze read data for specific targets and provides a detailed report output file. The ASAP bioinformatics method is described in U.S. Publication No. 2018/0173843, the contents of which are incorporated herein by reference.
- (3) FIGS. 3A-3E. An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific for *B. anthracis*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.
- (4) FIGS. 4A-4D An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific for *Y. pestis*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.
- (5) FIGS. 5A-5E An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific for *F. tularensis*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.
- (6) FIGS. 6A-6C An ASAP output result table, showing a subset of results from a large diverse sample set (693 isolates). The ASAP output demonstrates a general presence/absence for each select agent isolate or near neighbor isolate tested using UI-AmpSeq assays specific *B. pseudomallei*, *B. mallei*, known virulence determinants, near neighbor (NN) species, and antimicrobial resistance (AMR) targets.
- (7) FIG. 7. Sensitivity and specificity results for *B. anthracis* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.
- (8) FIGS. 8A-8C. Sensitivity and specificity results for *B. pseudomallei/mallei* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.
- (9) FIG. 9. Sensitivity and specificity results for *Y. pestis* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.
- (10) FIG. 10. Sensitivity and specificity results for *F. tularensis* assay, including amplicon yield (locus read count % of total sample library read count) for representative target and NN species.
- (11) FIG. 11 An Example Clinical Plate layout. The black wells are samples.

DETAILED DESCRIPTION

Definitions

(12) As used herein, the verb “comprise” as is used in this description and in the claims and its conjugations are used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded.

(13) As used herein, reference to an element by the indefinite article “a” or “an” does not exclude the possibility that more than one of the elements are present, unless the context clearly requires that there is one and only one of the elements. The indefinite article “a” or “an” thus usually means “at least one.”

(14) The sample in this method is preferably a biological sample from a subject. The term “sample” or “biological sample” or “environmental sample” is used in its broadest sense. Depending upon the embodiment of the invention, for example, a sample may comprise a bodily fluid including whole blood, serum, plasma, urine, saliva, cerebral spinal fluid, semen, vaginal fluid, pulmonary fluid, tears, perspiration, mucus and the like; an extract from a cell, chromosome, organelle, or membrane isolated from a cell; a cell; genomic DNA, RNA, or cDNA, in solution or bound to a substrate; a tissue; a tissue print, or any other material isolated in whole or in part from a living subject or organism. Biological samples may also include sections of tissues such as biopsy and autopsy samples, and frozen sections taken for histologic purposes such as blood, plasma, serum, sputum, stool, tears, mucus, hair, skin, and the like. Biological samples also include explants and primary and/or transformed cell cultures derived from patient tissues. In some embodiments, sample may comprise a portion of a non-animal organism, such as a plant (e.g., castor beans or derivatives thereof). In other embodiments, the sample comprises soil, water, air or other environmental material.

(15) In some embodiments, sample or biological sample may include a bodily tissue, fluid, or any other specimen that may be obtained from a living organism that may comprise additional living organisms. By way of example only, in some embodiments, sample or biological sample may include a specimen from a first organism (e.g., a human) that may further comprise an additional organism (e.g., bacteria, including pathogenic or non-pathogenic/commensal bacteria, viruses, parasites, fungi, including pathogenic or non-pathogenic fungi, etc.). In some embodiments, the additional organism may be separately cultured after isolation of the sample to provide additional starting materials for downstream analyses, in some embodiments, the sample or biological sample may comprise a direct portion of the additional, non-human organism and the host organism (e.g., a biopsy or sputum sample that contains human cells and bacteria).

(16) Some embodiments of the invention may comprise a multiplex assay. As used herein, the term “multiplex” refers to the production of more than one amplicon, PCR product, PCR fragment, amplification product, etc. in a single reaction vessel. In other words, multiplex is to be construed as the amplification of more than one target-specific sequences within a PCR reaction or assay within the same PCR assay mixture (e.g., more than one amplicon is produced within a single vessel that contains all of the reagents necessary to perform a PCR reaction). In some embodiments, a step prior to performing the PCR (or RT-PCR, quantitative RT-PCR, etc.) reaction can occur such that sets of primers and/or primers and probes are designed, produced, and optimized within a given set of reaction conditions to ensure proper amplicon production during the performance of the PCR.

(17) BioThreatSeq

(18) Currently, there are two approaches being used to identifying specimens in the environment. The first approach, metagenomics, tries to sequence “all” of the DNA in a sample and then to unravel its content computationally. Sequencing “all” of the DNA is difficult, slow, and very expensive. The computational approaches are improving but still contain many flaws that lead to false conclusions. A recent sensationalized report of DNA from anthrax and plague bacteria in the NYC subways illustrates the pitfalls of such endeavors (Mason et al., Cell Systems). Significant amounts of data are produced with metagenomics but frequently not enough for informative

signatures to differentiate pathogens from near neighbors. Deep sequencing of single specimens may cost nearly \$1,000 and take several weeks to generate. It is clearly not ready at this time for implementation for biosurveillance.

(19) The second approach, amplicon sequencing, involves the deep (>5,000×) sequencing of the 16S gene PCR amplicon to identify individual components of mixed bacterial communities. While the PCR primers are not specific, the intervening sequences can be highly informative and can be used to discriminate among bacterial taxa. Unfortunately for biothreat detection, the 16S gene has insufficient discrimination power to differentiate biothreat pathogens from their near neighbors. Discrimination power is a function of gene diversity and the 16S has low or no diversity among closely related bacteria. It cannot effectively identify a biothreat agent or distinguish from near-neighbor species.

(20) Increasing the amplicon discrimination power can be accomplished through comparative genomic analysis to identify diverse genomics regions. This is most effective when large genome databases are available and can be highly predictive of success once implemented. In clinical diagnostics, this approach is being used to identify multiple pathogens and to predict their virulence and resistance to antibiotics. PCR primers specific to a pathogen genera or species is sufficient, if there is additional DNA sequence information that can be leveraged for precise agent identification. Multiplex systems of several hundred amplicons are becoming common and provide coverage for dozens of pathogens. Sample preparation that works for real-time-PCR also works for this technology, so currently sampling schemes would adapt well to this type of assay. A multiple amplicon sequencing system would be easily adapted to changing targets with addition of new amplicons. Because of the multiplex nature of the assay, redundant amplicons can easily be included to verify the identification of a biothreat agent and even provide a differential identification of a near-neighbor species. Variation within the amplicons can be analyzed to identify drug resistance, virulence factors and subtype to the strain level.

(21) The ideal multiple amplicon sequencing system for identifying major biothreat agents should distinguish between the biothreat agent and its near-neighbor species using both amplification positive/negative criteria and qualitative analysis of sequence within the amplicons. This latter analysis provides strain identification and drug susceptibility identification. The analytical system should be supported by an automated interpretive software that generates actionable reports. Such a system includes quality assurance data to identify sample and/or process issues rapidly, to limit the effect of QC issues on final results.

(22) “BioThreatSeq” detects a presence, an absence, and/or a clinically important characteristic of nucleic acids from one or more microbial pathogens. BioThreatSeq is based upon very discriminating genetic regions bioinformatically identified using public and private genome sequences from microbial pathogens including but not limited to *Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, *Yersinia pestis* and Near Neighbor (NN) species. BioThreatSeq can be used to screen environmental samples for presence of target agent DNA, as well as war fighter and civilian patients for target agent carriage in the event of suspected exposure.

(23) In an embodiment, BioThreatSeq comprises a highly multiplexed amplicon sequencing assay. The assay is a highly informative screening tool capable of simultaneously detecting a presence of a microbial pathogen and a clinically important characteristic of the microbial pathogen without a live culturing step. Non-limiting examples of the clinically important characteristics include: virulence, or antibiotic resistance genetic signatures, etc.

(24) The utility of this assay has been demonstrated on several complex environmental and clinical specimen types including urine, wound swabs, sputum, air, soil, water samples. The superiority of BioThreatSeq over traditional typing techniques include high sensitivity (e.g., a low limit of detection) and high specificity (e.g., discriminating among strains, detecting antimicrobial resistance, and profiling virulence signatures, etc.) in target agent detection. BioThreatSeq is also

highly adaptable to new content, which allows for the flexibility to detect new biothreats agents and signatures. Thus, the assay methodology allows for the expansion of this tool to be used for several other BioThreat agents or applications.

(25) Target Specific Amplicon

(26) The Inventors used comparative genomics to identify a first genomic region that differentiates a target agent from its near neighbor relatives. In the first scenario, the first genomic region is present in all known target strains, and a lack of the first genomic region indicates an absence of the target agent in the sample. In the second scenario, the first genomic region not only is present in all known target strains, but is also absent in near-neighbor species. Thus, a presence of the first genomic region indicates a presence of the target agent in the sample.

(27) In certain non-limiting embodiments, the presence or absence of the first genomic region in the nucleic acids of the sample is determined by PCR using a first forward primer and a first reverse primer. The first forward primer and the first reverse primer amplify a Target Specific Amplicon, i.e., all strains of the threat agent, but not the near neighbors.

(28) Differential Target Amplicons

(29) The Inventors also used comparative genomics to identify a second genomic region that differentiates a target agent from its near neighbor relatives. The second genomic region is present in all strains of the near-neighbor relatives, but will not be in the target. Thus, a presence of the second genomic region indicates an absence of the target agent in the sample.

(30) In certain non-limiting embodiments, the presence or absence of the second genomic region in the nucleic acids of the sample is determined by PCR using a second forward primer and a second reverse primer. The second forward primer and second reverse primer amplify Differential Target Amplicons, i.e., all strains of the near neighbors, but not the threat agent. Differential identification assays can be included in the multiplex assay to help nullify any false positive results. This optional step offers interpretive value in complex species.

(31) The Inventors determined the exclusivity and inclusivity of the first forward and the first reverse primers in silico across all available threat agents (*Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, and *Yersinia pestis*) and near neighbor genomes. The in-silico validation included genomes from common contaminants such as humans. Because a large number of genomics sequences exist for both target and non-target organisms, the in-silico validation step eliminates any primers that are non-exclusive to the biothreat target. The assay primers were tested against the target and near-neighbor DNA templates to validate them under actual assay conditions.

(32) Primer Design

(33) The design of the first forward primer, the first reverse primer, the second forward primer, and the second reverse primer is consistent with a standard PCR method but is amendable to analysis using next-generation sequencing methods. This requirement includes the addition of “barcodes” to allow for indexing of samples for combining into single DNA sequencing batches. The technical details are provided in the PCT Patent Application entitled “Systems And Methods for Universal Tail-Based Indexing Strategies for Amplicon Sequencing” (International Application Number: PCT/US2014/064890; International Publication Number: WO 2015/070187 A2), the contents of which are hereby incorporated in their entirety.

(34) The Inventors determined the exclusivity and inclusivity of the second forward and the second reverse primers in silico across all available threat agents (*Bacillus anthracis*, *Burkholderia pseudomallei*, *Burkholderia mallei*, *Francisella tularensis*, and *Yersinia pestis*) and near neighbor genomes. The in-silico validation included genomes from common contaminants such as humans and soil DNA. Because a large number of genomics sequences exist for both target and non-target organisms, the in-silico validation step eliminates any primers that are non-exclusive to the near-neighbor species. The assay primers were tested against the target and near-neighbor DNA templates to validate them under actual assay conditions.

(35) In the third scenario, the first genomic region is present in all known target strains and at least one near-neighbor species. In this case, producing an exclusive amplicon is not feasible and the combination of amplification and internal sequence is needed to distinguish target from near-neighbors. In the absence of exclusive-target amplification, the amplicon sequence could provide definitive identification of the target and non-target agents.

(36) The Inventors has defined the phylogenetic structure of the first genomic region that includes both the target agent and its near neighbors and identified a variable internal sequence region which allows for: (1) differentiation of near neighbor from target species, (2) strain identification, (3) drug susceptibility identification, and/or (4) virulence prediction.

(37) Performance of the Multiplex Assays

(38) The Inventors have developed combined multi-agent amplicon sequencing assays for 2, 3, 4, 5, 6, or 7 biothreat agents and validated them under laboratory conditions. For the combined biothreat agent assays, important test parameters such as linearity, LOD, sensitivity, specificity, quantitative performance (absolute and relative), contaminant interference, performance with environmental samples (spikes), etc. have been determined.

(39) Software

(40) The Inventors have developed software that analyzes *B. anthracis* amplicon sequence data and provides actionable information (i.e., agent presence with confidence metrics, presence of virulence and antibiotic resistance factors, phylogenetic classification, etc.). The Inventors have also developed software that analyzes *B. anthracis* and other target agent (*F. tularensis*, *Y. pestis*, *B. mallei*, *B. pseudomallei*, *Brucella melitensis*, and *B. abortus*) and allow for on-site and remote reporting.

(41) BTSeq comprises target agent and near neighbor (NN) species identification assays, antimicrobial resistance (AMR) assays, virulence gene assays, and uses TGen North's amplicon sequencing analysis pipeline (ASAP) to report results.

(42) Use of the disclosed amplicon sequencing tool can be used to screen environmental samples for presence of target agent DNA, as well as war fighter and civilian patients for target agent carriage in the event of suspected exposure.

(43) In some embodiments, the present invention relates to a method of detecting *Bacillus anthracis* in a sample, comprising detecting at least one *B. anthracis*-specific amplicon selected from the group consisting of: CP008853.1_5309, CP008853.1_5316, CP012725.1_3629, CP012725.1_5103, CP012725.1_5107, JSZQ01000034.1_220, JSZS01000036.1_5, LGCC01000010.1_232, and LGCC01000048.1_280 in the sample, wherein the presence of the *B. anthracis*-specific amplicon indicates the presence of *B. anthracis* in the sample, and an absence of the *B. anthracis*-specific amplicon indicates an absence of *B. anthracis* in the sample.

(44) In other embodiments, the disclosed methods further comprise confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific amplicon selected from the group consisting of: NN_LOMU01000090.1_49, NN_LOQC01000013.1_3, and ChimpKiller_9-159 in the sample, wherein detecting the *B. anthracis* Near Neighbor-specific amplicon confirms the absence of *B. anthracis*.

(45) In yet other embodiments, the disclosed methods further comprise characterizing and/or subtyping *B. anthracis* by detecting at least one amplicon, single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: ChimpKiller_91-320, ChimpKiller_481-698, plcR, pagA, pX01, pX01, gyrA, parC, gyrB, rpoB, AA_2502, AA_2503, Ba_AmesAnc_4669915, Ba_AmesAnc_4001578, Ba_AmesAnc_1069024, Ba_AmesAnc_3668548, Ba_AmesAnc_371913, and Ba_AmesAnc_999035 in the sample.

(46) In certain aspects, the disclosed methods further comprise characterizing and/or subtyping *B. anthracis* by detecting at least one amplicon, single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: ChimpKiller_91-320, ChimpKiller_481-698, plcR, pagA, pX01, pX01, gyrA, parC, gyrB, rpoB, AA_2502, AA_2503, Ba_AmesAnc_4669915,

Ba_AmesAnc_4001578, Ba_AmesAnc_1069024, Ba_AmesAnc_3668548, Ba_AmesAnc_371913, and Ba_AmesAnc_999035 in the sample.

(47) In other aspects, the present invention relates to a method of detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in a sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon selected from the group consisting of: LWWC01000187.1_18, LWWB01000125.1_17183_17602, LXAY01000367.1_0_640, LWVY01000190.1_17226_17689, and LXAD01000059.1_24760_25075, wherein the presence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

(48) In some embodiments, the present invention provides a method of detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in the sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon selected from the group consisting of: LWWC01000187.1_18, LWWB01000125.1_17183_17602, LXAY01000367.1_0_640, LWVY01000190.1_17226_17689, and LXAD01000059.1_24760_25075, wherein the presence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

(49) In other embodiments, the present invention provides a method of detecting *B. pseudomallei* in a sample by detecting at least one *B. pseudomallei*-specific amplicon selected from the group consisting of: TTS1 BPSS1407, LXCC01000141.1_39296_39817, LXBY01000087.1_75760_76751, LXCD01000002.1_99652_100245, and LXCE01000123.1_34220_34747, wherein the presence of the *B. pseudomallei*-specific amplicon indicates the presence of *B. pseudomallei* in the sample, and an absence of the *B. pseudomallei*-specific amplicon indicates an absence of *B. pseudomallei* in the sample.

(50) In yet other embodiments, the present invention provides a method of detecting *B. mallei* in the sample by detecting at least one *B. mallei*-specific amplicon selected from the group consisting of: Bm 11589 and Bm 11767, wherein the presence of the *B. mallei*-specific amplicon indicates the presence of *B. mallei* in the sample, and an absence of the *B. mallei*-specific amplicon indicates an absence of *B. mallei* in the sample.

(51) In certain aspects, the disclosed methods further comprise characterizing and/or subtyping *B. pseudomallei* and/or *B. mallei* by detecting at least one one amplicon, single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: K9penA378-529, K9penA575-761, K9penA949-1172, pbp3-1, and pbp3-2 in the sample.

(52) In other aspects, the disclosed methods further comprise confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific single nucleotide polymorphism (SNP) or sequence variant (SV) selected from the group consisting of: NC 006350 2289827, NC 006350 133027, NC 006350 2248145-2248193, and NC 006350 988041-988089 in the sample, wherein detecting the *B. pseudomallei* or *B. mallei* Near Neighbor-specific single nucleotide polymorphism (SNP) or sequence variant (SV) confirms the absence of *B. pseudomallei* and *B. mallei*.

(53) In yet other aspects, the present invention provides a method of detecting *Francisella tularensis* in a sample by detecting at least one *F. tularensis*-specific amplicon selected from the group consisting of: *F. tularensis*_CP000915.1_1782, *F. tularensis*_CP000915.1-731, and Ft_dup_CP000915.1_197, wherein the presence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is present in the sample, and an absence of the *F. tularensis*-specific amplicon indicates that *F. tularensis* is absent in the sample.

(54) In some aspects, the disclosed methods further comprise confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific amplicon selected from the group consisting of: *F. tnoviciida*_CP009607.1, *F. philom*_CP009444.1_569, and *F.*

philom_CP009444.1_285 in the sample, wherein detecting the *F. tularensis* Near Neighbor-specific amplicon confirms the absence of *F. tularensis*.

(55) In other aspects, the disclosed methods further comprise confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV selected from the group consisting of: FtA1, FtA2, FtB, FtA, FtLVS_AM233362_1646546, FtLVS_AM233362_1643765, and FtLVS_AM233362_1562618 in the sample, wherein detecting the *F. tularensis* Near Neighbor-specific polymorphism confirms the absence of *F. tularensis*.

(56) In yet other aspects, the present invention provides a method of detecting *Yersinia pestis* in the sample by detecting at least one *Y. pestis*-specific amplicon selected from the group consisting of: *Y. pestis*_LPQY01000176.1_7, AGJT01000065.1_0_338, and FAUR01000053.1_96407_96884, wherein the presence of the *Y. pestis*-specific amplicon indicates the presence of *Y. pestis* in the sample, and an absence of the *Y. pestis*-specific amplicon indicates an absence of *Y. pestis* in the sample.

(57) In one embodiment, the disclosed methods further comprise confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV selected from the group consisting of: YpCO92_NC_003143_113190, YpCO92_NC_003143_161621, YpCO92_NC_003143_152213, YpCO92_NC_003143_129539, YpCO92_NC_003143_91203, YpCO92_NC_003143_121812, and Yp_AL590842.1_RX_SNP in the sample, wherein detecting the *Y. pestis* Near Neighbor-specific SNP or SV confirms the absence of *Y. pestis*.

(58) In another embodiment, the disclosed methods further comprise characterizing and/or subtyping *Y. pestis* by detecting at least one amplicon selected from the group consisting of: YpPGM_AL031866.1_81, YpPGM_31-205, Yp-p1202_42780-43194, Yp-p1202_126386-126750, and Yp-p1202_156402-156711 in the sample.

(59) In some embodiments, the presence or absence of *B. anthracis* in a sample is detected by identifying a specific mutation in the PlcR gene, a single base change at position 640, a nonsense mutation, which creates a dysfunctional protein. In other embodiments, the presence or absence of *B. anthracis* in a sample is detected by identifying the pXO1 and/or pXO2 plasmids.

(60) PlcR is a global transcriptional regulator which controls most of the secreted virulence factors in *B. cereus* and *B. thuringiensis*. It is chromosomally encoded and is ubiquitous throughout the cell (Agaisse, H. et al. (June 1999). "PlcR is a pleiotropic regulator of extracellular virulence factor gene expression in *Bacillus thuringiensis*". Molecular Microbiology. 32 (5): 1043-53). In *B. anthracis*, however, the plcR gene contains a single base change at position 640, a nonsense mutation, which creates a dysfunctional protein. While 1% of the *B. cereus* group carries an inactivated plcR gene, none of them carries the specific mutation found only in *B. anthracis* (Slamti, L. et al. (June 2004). "Distinct mutations in PlcR explain why some strains of the *Bacillus cereus* group are nonhemolytic". Journal of Bacteriology. 186 (11): 3531-8).

(61) The lack of PlcR in *B. anthracis* is a principle characteristic differentiating it from other members of the *B. cereus* group. While *B. cereus* and *B. thuringiensis* depend on the plcR gene for expression of their virulence factors, *B. anthracis* relies on the pXO1 and pXO2 plasmids for its virulence (Kolsto, A. et al. (October 2009). "What Sets *Bacillus anthracis* Apart from Other *Bacillus* Species?" Annual Review of Microbiology. 63 (1): 451-476). *Bacillus cereus* biovar anthracis, i.e. *B. cereus* with the two plasmids, is also capable of causing anthrax.

(62) In various embodiments, the disclosed methods identify an antibiotic resistance gene selected from a beta-lactamase gene, such as blaOXA, encoding extended spectrum OM class D beta-lactamases, blaCTX-M 82, blaCFX A4, encoding extended spectrum class A serine beta-lactamases, and AmpC, encoding the extended spectrum cephalosporin-resistant class C beta-lactamases; a multidrug efflux transporter system gene such as acrE, encoding a component of the AcrEF-ToIC multidrug efflux transporter system (Lau and Zgurskaya, 2005, J. Bacteriol. 187:7815); baeR, encoding a response regulator of the MdtABC multidrug efflux transporter system (Nagakubo et al., 2002, J. Bacteriol. 184:4161); emrY, encoding a component of the

EmrKY-ToIC multidrug efflux transporter system (Tanabe et al., 1997, J. Gen. Appl. Microbiol. 43:257); mdtD, encoding a component of the MdtABC multidrug efflux transporter system (Nagakubo et al., 2002, J. Bacteriol. 184:4161); and mdtN, encoding a multidrug resistance efflux pump from the major facilitator superfamily (Sulavik et al., 2001, Antimicrob. Agents Chemother. 45:1126); pbp2, encoding penicillin binding protein 2 (Bharat et al., 2015, Antimicrob. Agents Chemother. 59:5003); pbp4, encoding penicillin binding protein 4 (Sun et al., 2014, PLoS One 9:e97202); and aminoglycoside_strA (Scholz et al., 1989, Gene 75:271) encodes an aminoglycoside phosphotransferase, and Tetracycline_tet39 (Agerso and Guardabassi, 2005, J. Antimicrob. Chemother. 55:566) encodes a component of a tetracycline efflux pump.

(63) Other antibiotic resistance genes are provided in the Antibiotic Resistance Genes Database (ARDB), see Nucl. Acids Res. (2009) 37 (suppl 1): D443-D447, the World Wide Web (www) at ardb.cbcb.umd.edu, Antimicrob. Agents Chemother. July 2013 vol. 57 no. 7 3348-3357, and the NCBI database (the World Wide Web (www) at ncbi.nlm.nih.gov), the entire contents of which are hereby incorporated by reference.

(64) In various embodiments, the antibiotic resistance gene is one or more of the genes shown below:

(65) Aminocoumarins:

(66) Aminocoumarin-resistant DNA topoisomerases Aminocoumarin-resistant GyrB, ParE, ParY

Aminoglycosides: Aminoglycoside acetyltransferases AAC(1), AAC(2), AAC(3), AAC(6')

Aminoglycoside nucleotidyltransferases ANT(2''), ANT(3''), ANT(4), ANT(6), ANT(9)

Aminoglycoside phosphotransferases APH(2''), APH(3''), APH(3'), APH(4), APH(6), APH(7''),

APH(9) 16S rRNA methyltransferases ArmA, RaitA, RntB, RniC, Sgrn

β -Lactams: Class A β -lactamases AER, BLA1, CTX-M, KPC, SHV, TEM, etc. Class B (metallo-) β -lactamases BlaB, CcrA, IMP, NDM, VIM, etc. Class C β -lactamases ACT, AmpC, CMY, LAT, PDC, etc. Class D β -lactamases OXA β -lactamase mecA (methicillin-resistant PBP2)

Mutant porin proteins conferring antibiotic resistance Antibiotic-resistant Omp36, OmpF, PIB (por)

Genes Modulating β -Lactam Resistance: bla (blaI, blaR1) and mec (mecI, mecR1) operons

Chloramphenicol: Chloramphenicol acetyltransferase (CAT) Chloramphenicol phosphotransferase

Ethambutol: Ethambutol-resistant arabinosyltransferase (ErnB)

Mupirocin: Mupirocin-resistant isoleucyl-tRNA synthetases

MupA, MupB Peptide Antibiotics: Integral membrane protein MprF

Phenicol: Cfr 23S rRNA methyltransferase

Rifampin: Rifampin ADP-ribosyltransferase (Arr) Rifampin glycosyltransferase Rifampin monooxygenase Rifampin phosphotransferase Rifampin resistance RNA polymerase-binding proteins

DnaA, RbpA Rifampin-resistant beta-subunit of RNA polymerase (RpoB)

Streptogramins: Cfr 23S rRNA methyltransferase Erm 23S rRNA methyltransferases ErmA, ErmB, Erm(31), etc. Streptogramin resistance ATP-binding cassette (ABC) efflux pumps Lsa, MsrA, Vga, VgaB Streptogramin Vgb lyase Vat acetyltransferase

Fluoroquinolones: Fluoroquinolone acetyltransferase Fluoroquinolone-resistant DNA topoisomerases Fluoroquinolone-resistant GyrA, GyrB, ParC Quinolone resistance protein (Qnr)

Fosfomycin: Fosfomycin phosphotransferases FomA, FomB, FosC Fosfomycin thiol transferases FosA, FosB, FosX

Glycopeptides: VanA, VanB, VanD, VanR, VanS, etc.

Lincosamides: Cfr 23S rRNA methyltransferase Erm 23S rRNA methyltransferases ErmA, ErmB, Erm(31), etc. Lincosamide nucleotidyltransferase (Lin)

Linezolid: Cfr 23S rRNA methyltransferase

Macrolides: Cfr 23S rRNA methyltransferase Erm 23S rRNA methyltransferases ErmA, ErmB, Erm(31), Macrolide esterases EreA, EreB Macrolide glycosyltransferases GimA, Mgt, Ole

Macrolide phosphotransferases (MPH) MPH(2')-I, MPH(2')-II Macrolide resistance efflux pumps MefA, MefE, Mef

Streptothricin: Streptothricin acetyltransferase (sat)

Sulfonamides: Sulfonamide-resistant dihydropteroate synthases Sul1, Sul2, Sul3, sulfonamide-resistant FolP

Tetracyclines: Mutant porin PIB (por) with reduced permeability Tetracycline inactivation enzyme TetX Tetracycline resistance major facilitator superfamily (MFS) efflux pumps TetA, TetB, TetC, Tet30, Tet31, etc. Tetracycline resistance ribosomal protection proteins TetM, TetO, TetQ, Tet32, Tet36, etc.

Efflux Pumps Conferring Antibiotic Resistance: ABC antibiotic efflux pump MacAR-TolC, MsbA, MsrA, VgaB, etc. MFS antibiotic efflux pump EmrD, EmrAB-TolC, NorB, GepA, etc. Multidrug and toxic compound extrusion (MATE) transporter MepA Resistance-nodulation-cell division (RND) efflux pump AdeABC, AcrD, MexAB-OprM, mtrCDE, etc. Small multidrug resistance (SMR) antibiotic efflux pump EmrE

Genes Modulating Antibiotic Efflux: adeR, acrR, baeSR, mexR, phoPQ, mtrR

Multidrug Resistance: plasmid pLP1202

(67) In certain aspects, the disclosed methods the sample is obtained from a subject and the method further comprises administering at least one antibiotic to the subject.

(68) In one aspect, the at least one antibiotic is a fluoroquinolone. Non-limiting fluoroquinolones for use as described herein include levofloxacin, ofloxacin, ciprofloxacin, enoxacin, gatifloxacin, gemifloxacin, grepafloxacin, besifloxacin, lomefloxacin, moxifloxacin, norfloxacin, ofloxacin, pefloxacin, sparfloxacin, garenoxacin, trovafloxacin, sitafloxacin, and DX-619.

(69) In another aspect, the at least one antibiotic is an aminoglycoside such as amikacin, gentamicin, kanamycin, neomycin, netilmicin, paromomycin, streptomycin, or tobramycin.

(70) In another aspect, the at least one antibiotic is a carbapenem such as ertapenem, imipenem, meropenem, or chloramphenicol.

(71) In another aspect, the at least one antibiotic is a glycopeptide such as vancomycin.

(72) In another aspect, the at least one antibiotic is a lincosamide such as clindamycin.

(73) In another aspect, the at least one antibiotic is a macrolide/ketolide such as azithromycin, clarithromycin, dirithromycin, erythromycin, or telithromycin.

(74) In another aspect, the at least one antibiotic is a cephalosporin such as (1st generation) cefadroxil, cefazolin, cephalexin, cephalothin, cephapirin, and cephadrine; or (2nd generation) cefaclor, cefamandole, cefonicid, cefotetan, cefoxitin, cefprozil, cefuroxime, and loracarbef, or (3rd generation) cefdinir, cefditoren, cefixime, cefoperazone, cefotaxime, cefpodoxime, ceftazidime, ceftibuten, ceftizoxime, and ceftriaxone, or (4th generation) cefepime.

(75) In another aspect, the at least one antibiotic is a monobactam such as aztreonam.

(76) In another aspect, the at least one antibiotic is a nitroimidazole such as metronidazole.

(77) In another aspect, the at least one antibiotic is an oxazolidinone such as linezolid.

(78) In another aspect, the at least one antibiotic is a penicillin such as amoxicillin, amoxicillin/clavulanate, ampicillin, ampicillin/sulbactam, bacampicillin, carbenicillin, cloxacillin, dicloxacillin, methicillin, mezlocillin, nafcillin, oxacillin, penicillin G, penicillin V, piperacillin, piperacillin/tazobactam, ticarcillin, or ticarcillin/clavulanate.

(79) In another aspect, the at least one antibiotic is a streptogramin such as quinupristin/dalfopristin.

(80) In another aspect, the at least one antibiotic is a tetracycline such as demeclocycline, doxycycline, minocycline, or tetracycline.

(81) In another aspect, the at least one antibiotic is a β -lactam such as a penicillin, cephalosporin, carbapenem, or monobactam.

(82) The at least one antibiotic may be a physiologically acceptable salt, prodrug, or combination of any one of the aforementioned antibiotics.

(83) The following examples are given for purely illustrative and non-limiting purposes of the present invention.

EXAMPLES

Example 1. Detecting the Presence of Nucleic Acids from *Burkholderia*

(84) This protocol describes procedures for: (1) PCR amplification of multiplexed *Burkholderia* targets; (2) Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer); (3) SequelPrep™ Normalization Plate Kit (ThermoFisher); and (4) Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter).

(85) Universal-tailed gene-specific primers are pooled together in a “primer mix” in amounts relative to each other to help reduce PCR bias. Make the primer mixture by combining the primers and 1×TE (e.g., according to Table 8). Vortex and spin down each primer stock. The primer mixture can be stored at -20° C. Plan and arrange the layout of where the samples will go on a 96-well plate. If clinical samples are being processed, make sure to space the samples on the plate accordingly. FIG. 11 can be used as an example clinical plate layout. Black wells are samples. Combine the following volumes of reagents as described in Table 10, except IPSC to create the *Burkholderia* Multiplex Master Mix. Reagents should be thawed and mixed before use. Vortexing PCR mastermix should be avoided to prevent damaging the enzymes in the mixture.

(86) To prepare *Burkholderia* Multiplex, on experimental day, thaw out an aliquot of Internal Plasmid Sequencing Control (IPSC) at 10⁶ copies per uL, dilute down to 10³ copies per uL by three serial dilutions of 1 to 10. Make dilutions in tubes for better vortexing & spinning. For single reaction no-template control (NTC) reaction, add 12.5 µL Q5 2× HotStart, 5 µL 5M Betaine, and 4.5 µL Diluted Primer Mix. For single reaction, add 12.5 µL Q5 2× HotStart, 5 µL 5M Betaine, 1 µL IPSC 1000 copies per µL, and 4.5 µL Diluted Primer Mix. To prepare for Master Mix reaction, add 2475 µL Q5 2× HotStart, 990 µL 5M Betaine, 198 µL IPSC 1000 copies per µL, and 891 µL Diluted Primer.

(87) Gently mix template DNA and spin down (Do not vortex genomic DNA). Add 2 µL of template DNA to its appropriate well, along with 2 µL H.sub.2O for IPSC, and 3 µL H.sub.2O for NTC. Seal plate with a thermocycler seal. Spin down the plate. Using a heated lid, put plate on thermocycler and run the following parameters:

(88) TABLE-US-00001 Step Repeats Temperature (° C.) Time initial denaturation 1 98 5 min denaturation 35 98 30 sec annealing 68 15 sec extension 72 20 sec final extension 1 72 2 min cooling 1 10 forever

(89) Spin plates (*Burkholderia* target plate and *Bacillus*, *Yersinia*, *Francisella* target plate) down once the thermocycler finishes.

(90) Prepare AMPure Beads. Prepare 10 mM Tris-HCl 0.05% Tween-20 in H.sub.2O by adding 400 µL 10 mM Tris-HCl, 20 µL 0.05% Tween-20, and 39.580 mL Molecular Grade H.sub.2O, and heat to 50° C. Equilibrate the bead to Room Temperature for 30 minutes. Add beads in a 1:1 ratio with reaction volume to each well (30 µL) and mix well by pipetting. Incubate the bead/reaction mixture for 5 minutes. Place the 96-well plate onto a magnetic stand, incubate for another 5 minutes. Aspirate supernatant out of wells without disturbing the beads. If beads were disturbed, let them incubate for another 2 minutes. Be sure to remove as much liquid as you can. Twice wash the beads by adding 80% EtOH (32 ml 100% Ethanol, and 8 ml Molecular Grade H.sub.2O) to completely cover beads (~200 µL) and incubate for 30 seconds. Aspirate. Fully remove liquids after the wash. Move plate off magnetic stand and allow beads to dry. Be sure to keep a close watch on the beads. If the beads start to crack, the DNA will be difficult to elute out. Move plate off the magnetic stand and add 32.54 heated 10 mM Tris-HCl 0.05% Tween-20 in H.sub.2O to the wells, mix well. Incubate for 2 minutes. Move plate to magnetic stand and incubate for 2 minutes. Remove 304 of supernatant and transfer it to a new well, do not disturb or transfer any beads.

(91) *Burkholderia*, *Bacillus*, *Francisella*, and *Yersinia* AmpSeq amplicons require two bead cleanups before Extension PCR. Repeat steps 12-21.

(92) Detecting the Presence of Nucleic Acids from *Bacillus anthracis*, *Yersinia*, and *Francisella* UT-AmpSeq PCR and Bead Cleanup.

(93) 1. PCR amplification of multiplexed *Bacillus*, *Yersinia*, and *Francisella* targets. Universal-tailed gene-specific primers are pooled together in a “primer mix” in amounts relative to each other to help reduce PCR bias. These amounts have been previously optimized. Please follow the Primer Mix parameters to create the needed mix for the multiplex currently in use. 2. Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer). Enter the total number of samples in the box below. Primer Mix Parameters, # of Samples 3. SequelPrep™ Normalization Plate Kit (ThermoFisher). 4. Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter). Protocol

SOP for *Burkholderia* UT-AmpSeq PCR and Bead Cleanup

(94) This SOP describes procedures for the following: 1. PCR amplification of multiplexed *Burkholderia* targets 2. Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer) 3. SequelPrep™ Normalization Plate Kit (ThermoFisher) 4. Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter)

Steps and Procedures 1. Universal-tailed gene-specific primers are pooled together in a “primer mix” in amounts relative to each other to help reduce PCR bias. These amounts have been previously optimized. Please follow the Primer Mix parameters to create the needed mix for the multiplex currently in use 2. Enter the total number of samples in the box below. Primer Mix Parameters # of Samples 180 ensure that all values in column K “How much starting primer conc. to add in mix stock” are above 2.0u1 and not highlighted in red 3a. Make the primer mixture by combining the following primers. 3b. Vortex and spin down each primer stock. 3c. Using the “Start (uM)” concentration primer stock of each primer, add the volume from “Amount to add (uL)” into a 1.7 mL microcentrifuge tube, unless “Total (uL)” at bottom of table is above 1200, then split volume evenly across necessary tubes. Vortex to mix and spin. (Refer to Table 8) 4a. This mixture can be stored at -20° C. for future use. To use mixture, let thaw, vortex and spin down. 4b. If using mixture previously made, write down the initials of the person who made it and when: Initials _____ Date _____ 5. Plan and arrange the layout of where your samples will go on a 96-well plate. If you are processing clinical samples make sure to space your samples on the plate accordingly. 6. Use the Plate Maps sheet for convenience and record keeping. (Refer to FIG. 11). 6a. Reagents should be thawed and mixed before use. Avoid vortexing PCR mastermix as this can damage the enzymes in the mixture. 6b. Combine the following volumes of reagents as described in the following table except IPSC to create the *Burkholderia* Multiplex Master Mix

(95) TABLE-US-00002 *Burkholderia* Multiplex uL for single uL addition Reagent reaction in Master Mix Q5 2x HotStart 12.5 2475 Betaine 5M 5 990 IPSC 1000 copies/uL 1 198 Diluted Primer Mix 4.5 891 4554 6c. Add 22 uL of *Burkholderia* Multiplex Master Mix without IPSC to any NTC reactions you are processing 7a. Thaw out an aliquot of Internal Plasmid Sequencing Control (IPSC) at 10^6 copies per uL. Dilute this down to 10^3 copies/uL by three serial dilutions of 1 to 10. Make dilutions in tubes for better vortexing & spinning Make this fresh the day of. 7b. Add volume with the # of NTCs subtracted of 10^3 copies/uL IPSC to master mix. For example, if you had 3 NTCs that you had aliquoted master mix for, and the above table indicated 9 uL of IPSC be added, you would add 6 instead. 7d. Mix well and spin down 7e. Add 23 uL of Master Mix to each appropriate well on your plate 8a. Gently mix template DNA and spin down (Do not vortex genomic DNA) 8b. Add 2 uL of template DNA to its appropriate well, along with 2 uL H.sub.2O for IPSC, and 3 uL H.sub.2O for NTC 8c. Seal plate with a thermocycler seal 8d. Spin down plate 9. Using a heated lid, put plate on thermocycler and run the following parameters

(96) TABLE-US-00003 Step: Reps: Temp: Time: initial denaturation 1 98 5 min denaturation 35 98 30 sec annealing 68 15 sec extension 72 20 sec final extension 1 72 2 min cooling 1 10 forever 10. During this time, take out AMPure Beads to equilibrate them to Room Temperature for 30 minutes

and heat some 10 mM Tris-HCl 0.05% Tween-20 in H.sub.2O to 50 C.

(97) TABLE-US-00004 10 mM Tris-HCl 0.05% Tween-20 in H₂O Example Formula 10 mM Tris-HCl 400 uL 0.05% Tween-20 20 uL Molecular Grade H₂O 39.580 mL 80% Ethanol in H₂O 100% Ethanol 32 mL MBG H₂O 8 mL 11. Spin plates (*Burkholderia* target plate and *Bacillus*, *Yersinia*, *Francisella* target plate) down once the thermocycler finishes 12. Combine 15 uL of *Burkholderia* target reaction with 15 uL of *Bacillus*, *Yersinia*, *Francisella* target reaction 13. Add beads in a 1:1 ratio with reaction volume to each well (30 uL) and mix well by pipette 14. Incubate the bead/reaction mixture for 5 minutes 15. Place 96-well plate onto a magnetic stand, incubate for another 5 minutes 16. Aspirate supernatant out of wells without disturbing the beads. If beads ARE disturbed, let them incubate for another 2 minutes. Be sure to remove as much liquid as you can 17a. Add 80% EtOH to completely cover beads (~200 uL) and incubate for 30 seconds. Aspirate. 17b. Repeat 16a and remove as much liquid as you can (two washes total), following with a 20 uL pipette to ensure full removal 18. Move plate off magnetic stand and allow beads to dry. Be sure to keep a close watch on the beads. If the beads start to crack, the DNA will be harder to elute out. 19. Move plate off the magnetic stand and add 32.5 uL heated 10 mM Tris-HCl 0.05% Tween-20 in H.sub.2O to the wells, mix well 20. Incubate for 2 minutes 21. Move plate to magnetic stand and incubate for 2 minutes 22. Remove 30 uL of supernatant and transfer it to a new well, do not disturb or transfer any beads 22. *Burkholderia*, *Bacillus*, *Francisella*, and *Yersinia* AmpSeq amplicons require two bead cleanups before Extension PCR. Repeat steps 12-21 23. Store amplicons at -20 C Index Extension PCR 24. Thaw, gently mix, and spin down the following reagents in the following amounts for the Index Extension of the Target Amplicons * Amounts are in respect to number of samples entered in Step 2

(98) TABLE-US-00005 Index Extension Master Mix Reagent uL Lot# 2x Kapa Hifi 2475 Betaine 5M 990 Molecular Grade H₂O 693 25. Combine the above volumes together, mix gently, and spin down. 26a. Each reaction will require a unique pair of index primers (UT1 and UT2), prepare a chart of what indexes will be used and where 26b. Thaw, vortex, and spin down the stock 10 uM aliquots of each index that will be used for this run 26c. If some tubes appear empty, create a new 10 uM aliquot of that index. Dilute in TE. 27. Once all indexes are accounted for, add 21 uL of Index Extension Master Mix to each appropriate well in a 96-well plate 28. Add 1 uL of each 10 uM index to its appropriate well 29a. After all UT1 and UT2 indexes have been added to their wells add 2 uL of CLEANED AMPLICONS 29b. The following should now be in each reaction well

(99) TABLE-US-00006 12.5 uL 2x Kapa Hifi 3.5 uL H₂O 5 uL 5M Betaine 2 uL DNA 1 uL 10 uM UT1 1 uL 10 uM UT2 25 uL 30. Seal the plate with a thermocycler seal 31. Spin down plate 32a. Using a heated lid, put plate on thermocycler and run the following parameters

(100) TABLE-US-00007 Step: Reps: Temp: Time: initial denaturation 1 98 2 min denaturation 8 98 30 sec annealing 60 20 sec extension 72 30 sec final extension 1 72 2 min cooling 1 10 forever 32b. After the PCR has completed, spin down plate 33a. Samples will be cleaned and normalized using the Invitrogen SequalPrep system 33b. In a new plate, add equal amounts of illex DNA template and SequalPrep Normalization Binding Buffer 33c. Mix completely by pipette mixing several times, take care not to etch the sides of the well with the pipette tip 33d. Incubate the plate for 1 hour at room temperature to allow binding of DNA to the plate surface (longer than 1 hr is acceptable but will not increase binding or final elution concentration, can be overnight) 33e. Aspirate the liquid from the wells 33f. Add 50 uL Sequal Prep Normalization Wash Buffer, mix by pipetting up and down twice 33g. Completely aspirate the buffer, a small amount of residual Wash Buffer (1-3 uL) is typical 33h. Add 20 uL SequalPrep Normalization Elution Buffer to each well of the plate, mix by pipette 33i. Incubate at room temperature for 5 minutes 33j. Transfer samples to a new plate 34a. Samples should all be normalized now so pool them together in equal volumes 34b. The final DNA concentration will be fairly low, so perform an AMPure XP bead cleanup on the pool at a 1:1 ratio of pool to beads (be sure to note total volume of pooled samples) 34c. However, when eluting the DNA off the beads with heated Tris-Tween use 1/10 the initial pool volume used

35. Store DNA at -20 C

SOP for *Bacillus*, *Yersinia*, and *Francisella* UT-AmpSeq PCR and Bead Cleanup

(101) This SOP describes procedures for the following: 1. PCR amplification of multiplexed *Bacillus*, *Yersinia*, and *Francisella* targets 2. Index extension PCR to prepare amplicons for MiSeq (ILLUMINA® sequencer) 3. SequelPrep™ Normalization Plate Kit (ThermoFisher) 4. Agencourt AMPure XP bead cleanup for PCR purification (Beckman Coulter)

Steps and Procedures 1. Universal-tailed gene-specific primers are pooled together in a “primer mix” in amounts relative to each other to help reduce PCR bias. These amounts have been previously optimized. Please Follow the Primer Mix Parameters to Create the Needed Mix for the Multiplex Currently in use 2. Enter the total number of samples in the box below. Primer Mix Parameters # of Samples 180 ensure that all values in column K “How much starting primer conc. to add in mix stock” are above 2.0u1 and not highlighted in red 3a. Make the primer mixture by combining the following primers. 3b. Vortex and spin down each primer stock. 3c. Using the “Start (uM)” concentration primer stock of each primer, add the volume from “Amount to add (uL)” into a 1.7 mL microcentrifuge tube, unless “Total (uL)” at bottom of table is above 1200, then split volume evenly across necessary tubes. Vortex to mix and spin. (Refer to Table 10) 4a. This mixture can be stored at -20° C. for future use. To use mixture, let thaw, vortex and spin down. 4b. If using mixture previously made, write down the initials of the person who made it and when:

Initials_____ Date_____ 5. Plan and arrange the layout of where your samples will go on a 96-well plate. If you are processing clinical samples make sure to space your samples on the plate accordingly. 6. Use the Plate Maps sheet for convenience and record keeping. (Refer to FIG. 11). 6a. Reagents should be thawed and mixed before use. Avoid vortexing PCR mastermix as this can damage the enzymes in the mixture. 6b. Combine the following volumes of reagents as described in the following table except IPSC to create the *Bacillus*, *Francisella*, and *Yersinia* Multiplex Master Mix

(102) TABLE-US-00008 uL *Bacillus*, *Francisella*, addition in *Yersinia* Multiplex uL for Master Reagent single reaction Mix Q5 2x HotStart 12.5 2475 H2O 5 990 IPSC 1000 copies/uL 1 198 Diluted Primer Mix 4.5 891 4554 6c. Add 22 uL of *Burkholderia* Multiplex Master Mix without IPSC to any NTC reactions you are processing 7a. Thaw out an aliquot of Internal Plasmid Sequencing Control (IPSC) at 10{circumflex over ()}6 copies per uL. Dilute this down to 10{circumflex over ()}2 copies/uL by three serial dilutions of 1 to 10. Make dilutions in tubes for better vortexing & spinning Make this fresh the day of. 7b. Add volume with the # of NTCs subtracted of 10{circumflex over ()}2 copies/uL IPSC to master mix. For example, if you had 3 NTCs that you had aliquoted master mix for, and the above table indicated 9 uL of IPSC be added, you would add 6 instead. 7d. Mix well and spin down 7e. Add 23 uL of Master Mix to each appropriate well on your plate 8a. Gently mix template DNA and spin down (Do not vortex genomic DNA) 8b. Add 2 uL of template DNA to its appropriate well, along with 2 uL H.sub.2O for IPSC, and 3 uL H.sub.2O for NTC 8c. Seal plate with a thermocycler seal 8d. Spin down plate 9. Using a heated lid, put plate on thermocycler and run the following parameters

(103) TABLE-US-00009 Step: Reps: Temp: Time: initial denaturation 1 98 5 min denaturation 35 98 30 sec annealing 55 15 sec extension 72 20 sec final extension 1 72 2 min cooling 1 10 forever 10. During this time, take out AMPure Beads to equilibrate them to Room Temperature for 30 minutes and heat some 10 mM Tris-HCl 0.05% Tween-20 in H.sub.2O to 50 C

(104) TABLE-US-00010 10 mM Tris-HCl 0.05% Tween-20 in H2O Example Formula 10 mM Tris-HCl 400 uL 0.05% Tween-20 20 uL Molecular Grade H2O 39.580 mL 80% Ethanol in H2O 100% Ethanol 32 mL MBG H2O 8 mL 11. Spin plates (*Burkholderia* target plate and *Bacillus*, *Yersinia*, *Francisella* target plate) down once the thermocycler finishes 12. Combine 15 uL of *Burkholderia* target reaction with 15 uL of *Bacillus*, *Yersinia*, *Francisella* target reaction 13. Add beads in a 1:1 ratio with reaction volume to each well (30 uL) and mix well by pipette 14. Incubate the bead/reaction mixture for 5 minutes 15. Place 96-well plate onto a magnetic stand, incubate for

another 5 minutes 16. Aspirate supernatant out of wells without disturbing the beads. If beads ARE disturbed, let them incubate for another 2 minutes. Be sure to remove as much liquid as you can 17a. Add 80% EtOH to completely cover beads (~200 uL) and incubate for 30 seconds. Aspirate. 17b. Repeat 16a and remove as much liquid as you can (two washes total), following with a 20 uL pipette to ensure full removal 18. Move plate off magnetic stand and allow beads to dry. Be sure to keep a close watch on the beads. If the beads start to crack, the DNA will be harder to elute out. 19. Move plate off the magnetic stand and add 32.5 uL heated 10 mM Tris-HCl 0.05% Tween-20 in H₂O to the wells, mix well 20. Incubate for 2 minutes 21. Move plate to magnetic stand and incubate for 2 minutes 22. Remove 30 uL of supernatant and transfer it to a new well, do not disturb or transfer any beads 22. *Burkholderia*, *Bacillus*, *Francisella*, and *Yersinia* AmpSeq amplicons require two bead cleanups before Extension PCR. Repeat steps 12-21 23. Store amplicons at -20 C Index Extension PCR 24. Thaw, gently mix, and spin down the following reagents in the following amounts for the Index Extension of the Target Amplicons *Amounts are in respect to number of samples entered in Step 2

(105) TABLE-US-00011 Index Extension Master Mix Reagent uL 2x Kapa Hifi 2475 Betaine 5M 990 Molecular Grade H₂O 693 4158 Lot# 25. Combine the above volumes together, mix gently, and spin down. 26a. Each reaction will require a unique pair of index primers (UT1 and UT2), prepare a chart of what indexes will be used and where 26b. Thaw, vortex, and spin down the stock 10 uM aliquots of each index that will be used for this run 26c. If some tubes appear empty, create a new 10 uM aliquot of that index. Dilute in TE. 27. Once all indexes are accounted for, add 21 uL of Index Extension Master Mix to each appropriate well in a 96-well plate 28. Add 1 uL of each 10 uM index to its appropriate well 29a. After all UT1 and UT2 indexes have been added to their wells add 2 uL of CLEANED AMPLICONS 29b. The following should now be in each reaction well

(106) TABLE-US-00012 12.5 uL 2x Kapa Hifi 3.5 uL H₂O 5 uL 5M Betaine 2 uL DNA 1 uL 10 uM UT1 1 uL 10 uM UT2 25 uL 30. Seal the plate with a thermocycler seal 31. Spin down plate 32. Using a heated lid, put plate on thermocycler and run the following parameters

(107) TABLE-US-00013 Step: Reps: Temp: Time: initial denaturation 1 98 2 min denaturation 8 98 30 sec annealing 60 20 sec extension 72 30 sec final extension 1 72 2 min cooling 1 10 forever 32b.

After the PCR has completed, spin down plate 33a. Samples will be cleaned and normalized using the Invitrogen SequalPrep system 33b. In a new plate, add equal amounts of illex DNA template and SequalPrep Normalization Binding Buffer 33c. Mix completely by pipette mixing several times, take care not to etch the sides of the well with the pipette tip 33d. Incubate the plate for 1 hour at room temperature to allow binding of DNA to the plate surface (longer than 1 hr is acceptable but will not increase binding or final elution concentration, can be overnight) 33e.

Aspirate the liquid from the wells 33f. Add 50 uL Sequal Prep Normalization Wash Buffer, mix by pipetting up and down twice 33g. Completely aspirate the buffer, a small amount of residual Wash Buffer (1-3 uL) is typical 33h. Add 20 uL SequalPrep Normalization Elution Buffer to each well of the plate, mix by pipette 33i. Incubate at room temperature for 5 minutes 33j. Transfer samples to a new plate 34a. Samples should all be normalized now so pool them together in equal volumes 34b. The final DNA concentration will be fairly low, so perform an AMPure XP bead cleanup on the pool at a 1:1 ratio of pool to beads (be sure to note total volume of pooled samples) 34c. However, when eluting the DNA off the beads with heated Tris-Tween use 1/10 the initial pool volume used 35. Store DNA at -20 C

Burkholderia pseudomallei, *Burkholderia mallei*, *Bacillus anthracis*, *Yersinia pestis* and *Francisella tularensis* are all Tier 1 select agents, posing a potentially severe threat to public health..sup.1 Current surveillance methods rely upon single locus PCR techniques that allow for only presence/absence of SA results. Has been known to lead to false positives, especially due to the complexity of environmental samples including huge numbers of microorganisms, many of which can be highly similar target pathogens. Constantly working to develop technologies that significantly improve the identification of biological contaminants in varying sample types Both sequencing costs and sequencing time are decreasing..sup.2 Targeted

amplicon sequencing can provide the necessary information at a fraction of the cost of WGS. Targeted amplicon sequencing can also provide a more manageable data set for researchers with less background in bioinformatics with an appropriate processing tool.

Summary With the cost of sequencing decreasing and the ability to combine higher and higher numbers of samples on a single run, amplicon sequencing provides a cost effective alternative to individual PCR methods. A screening panel of target organism strains as well as near neighbor strains allowed for differentiation with 100% sensitivity for all target agents and 91-100% specificity.

Future Directions Limit of detection testing across a subset of target organisms in a pristine sample. Limit of detection testing across a subset of target organisms in samples with environmental and human backgrounds. Testing on other sequencing platforms.

REFERENCES

- (108) .sup.1 Select Agents and Toxins Regulations. 42 C.F.R. Part 73.3 HHS Select Agents and Toxins .sup.2 Goodwin S, McPherson J D, McCombie W R. Coming of age: ten years of next-generation sequencing technologies. Nat Rev Genet. 2016
- (109) TABLE-US-00014 TABLE 1 BTseq Loci Summary *B. pseudo- B. mallei/ F. Y anthracis mallei tularensis pestis* Target Species Loci PA 9 12* 3 3 NN Species Loci PA 4 0 3 0 NN Species Loci SNP/SV 7 9 7 7 AMR Loci 6 4 0 3 Virulence/Plasmid Loci 4 0 0 2 Total 30 25 13 15
- *Five loci amplify both Bp and Bm
- (110) TABLE-US-00015 TABLE 2 SNPs for detecting the presence of nucleic acids from *Bacillus anthracis*. Ba_AmesAnc LocusID 1069024::91 1712462::85 3668548::88 371913::77 388555::108 3981642::97 4001578::113 Reference T C C T A A C Ba_A0098_S45_L001 T C C T A A C Ba_A0330_S46_L001 T C C T A A C Ba_A0490_S47_L001 T C C T A A C Ba_A0605_S48_L001 T C C T A A C Ba_A0615_S49_L001 T C C T A A C Ba_A0706_S50_L001 T C C T A A C Ba_A071_S52_L001 T C C T A A C Ba_A072_S58_L001 T C C T A A C Ba_A073_S59_L001 T C C T A A C Ba_A074_S60_L001 T C C T A A C Ba_A0767_S51_L001 T C C T A A C Ba_A0847_S53_L001 T C C T A A C Ba_A1085_S54_L001 T C C T A A C Ba_A2010_S55_L001 T C C T A A C Ba_A2105_S56_L001 T C C T A A C Ba_A2175_S57_L001 T C C T A A C Ba_NN295171- A X X X X X T grainy_S63_L001 Ba_NN295171- X X X X X X X smooth_S70_L001 Ba_NN295618_S71_L001 X X X X X X X Ba_NNAI-hakem- X X X X X X X grainy_S77_L001 Ba_NNAI-hakem- A X X C G G T smooth_S64_L001 Ba_NNATCC10792_S62_L001 A X X C G G T Ba_NNATCC13402_S76_L001 X X X X X X X Ba_NNATCC31293_S74_L001 X X X X X X T Ba_NNBacillus- X X X X X X X cereus_S69_L001 Ba_NNBacillus- A X X X X X T megaterium_S75_L001 Ba_NNFRI33_S72_L001 X X X X X X X Ba_NNFRI35_S73_L001 A X X X G G T Ba_NNHD- A X T C G G T 1011_S93_L001 Ba_NNHD- A X X C X X T 1012_S94_L001 Ba_NNHD- A X X X X G T 1015_S61_L001 Ba_NNHD- A X X C G G T 288_S81_L001 Ba_NNHD- X X X X X X T 34_S78_L001 Ba_NNHD-44- X X X C X G T grainy_S79_L001 Ba_NNHD-44- X X X X X X X smooth_—S66_L001 Ba_NNHD- A X X C G X T 47_S80_L001 Ba_NNHD-526- A X X T X G T grainy_S82_L001 Ba_NNHD-526- A T X C G G T smooth_—S67_L001 Ba_NNHD- X X X X X X T 557_S83_L001 Ba_NNHD-571- A X T C G G T grainy_S84_L001 Ba_NNHD-571- A X T C G G T smooth_S65_L001 Ba_NNHD- X X X X X X T 621_S85_L001 Ba_NNHD- A X X X G X T 681_S86_L001 Ba_NNHD- X X X X X X X 682_S87_L001 Ba_NNHD- X X X X X X X 711_S88_L001 Ba_NNHD- A X X X X X T 754_S89_L001 Ba_NNHD- X X X X X X X 789_S90_L001 Ba_NNHD- A X X X G X T 930_S91_L001 Ba_NNHD-974- A X X X G G T grainy_S92_L001 Ba_NNHD-974- A X X X G G T smooth_S68_L001 Ba_NNNTC_Ba_1_S95_L001 X X X X X X X Ba_NNNTC_Ba_2_S96_L001 X X X X X X X F1057_S97_L001 X X X X X X X F1062_S98_L001 X X X X X X X NTC_Ft1_S99_L001 X X X X X X X NTC_Ft2_S100_L001 X X X X X X X NTC_Yp1_55_S104_L001 X X X X X X X NTC_Yp2_55_S105_L001 X X X X X X C NTC_Yp3_60_S109_L001 X X X X X X C

NTC_Yp4_60_S100_L001 X X X X X X C NTC_Yp5_65_S114_L001 X X X X X X C
 NTC_Yp6_65_S115_L001 X X X X X X C Yp1763_55_S101_L001 X X X X X X X
 Yp1763_60_S106_L00 X X X X X X X Yp1763_65_S111_L001 X X X X X X X
 Yp2051_55_S102_L001 X X X X X X X Yp2051_60_S107_L001 X X X X X X X
 Yp2051_65_S112_L001 X X X X X X X Yp2126_55_S103_L001 X X X X X X X
 Yp2126_60_S108_L001 X X X X X X X Yp2126_65_S113_L001 X X X X X X X Ba_AmesAnc
 LocusID 4087624::116 4669915::40 734209::105 999035::79 plcR::147 Reference G T T G A
 Ba_A0098_S45_L001 G T T G A Ba_A0330_S46_L001 G T T G A Ba_A0490_S47_L001 G T T
 G A Ba_A0605_S48_L001 G T T G A Ba_A0615_S49_L001 G T T G A Ba_A0706_S50_L001 G
 T T G A Ba_A071_S52_L001 G T T G A Ba_A072_S58_L001 G T T G A Ba_A073_S59_L001 G
 T T G A Ba_A074_S60_L001 G T T G A Ba_A0767_S51_L001 G T T G A Ba_A0847_S53_L001
 G T T G A Ba_A1085_S54_L001 G T T G A Ba_A2010_S55_L001 G T T G A
 Ba_A2105_S56_L001 G T T G A Ba_A2175_S57_L001 G T T G A Ba_NN295171- X X X A X
 grainy_S63_L001 Ba_NN295171- X X X X X smooth_S70_L001 Ba_NN295618_S71_L001 X X
 X X X Ba_NNAI-hakem- X X X X X grainy_S77_L001 Ba_NNAI-hakem- T C C A X
 smooth_S64_L001 Ba_NNATCC10792_S62_L001 T C C A C Ba_NNATCC13402_S76_L001 X
 X X X X Ba_NNATCC31293_S74_L001 X X X X X Ba_NNBacillus- X X X X X
 cereus_S69_L001 Ba_NNBacillus- X X X X X megaterium_S75_L001 Ba_NNFRI33_S72_L001
 X X X X X Ba_NNFRI35_S73_L001 T C X A C Ba_NNHD- T C C A C 1011_S93_L001
 Ba_NNHD- T C X A C 1012_S94_L001 Ba_NNHD- X C X A X 1015_S61_L001 Ba_NNHD- X C
 C A X 288_S81_L001 Ba_NNHD- X X X X X 34_S78_L001 Ba_NNHD-44- X C X X C
 grainy_S79_L001 Ba_NNHD-44- X X X X X smooth_—S66_L001 Ba_NNHD- X C X A X
 47_S80_L001 Ba_NNHD-526- X C X X A grainy_S82_L001 Ba_NNHD-526- T C X A X smooth_
 —S67_L001 Ba_NNHD- X X X X X 557_S83_L001 Ba_NNHD-571- T C C A C
 grainy_S84_L001 Ba_NNHD-571- T C C A C smooth_S65_L001 Ba_NNHD- X X X X X
 621_S85_L001 Ba_NNHD- X C X A X 681_S86_L001 Ba_NNHD- X X X X X 682_S87_L001
 Ba_NNHD- X X X X X 711_S88_L001 Ba_NNHD- X C X A X 754_S89_L001 Ba_NNHD- X X
 X X X 789_S90_L001 Ba_NNHD- X C X A X 930_S91_L001 Ba_NNHD-974- X C X X X
 grainy_S92_L001 Ba_NNHD-974- X C X X X smooth_S68_L001 Ba_NNNTC_Ba_1_S95_L001
 X X X X X Ba_NNNTC_Ba_2_S96_L001 X X X X X F1057_S97_L001 X X X X X
 F1062_S98_L001 X X X X X NTC_Ft1_S99_L001 X X X X X NTC_Ft2_S100_L001 X X X X X
 NTC_Yp1_55_S104_L001 X X X X X NTC_Yp2_55_S105_L001 X X X X A
 NTC_Yp3_60_S109_L001 X X X X X NTC_Yp4_60_S110_L001 X X X X A
 NTC_Yp5_65_S114_L001 X X X X X NTC_Yp6_65_S115_L001 X X X X A
 Yp1763_55_S101_L001 X X X X X Yp1763_60_S106_L00 X X X X X Yp1763_65_S111_L001
 X X X X X Yp2051_55_S102_L001 X X X X X Yp2051_60_S107_L001 X X X X X
 Yp2051_65_S112_L001 X X X X X Yp2126_55_S103_L001 X X X X X Yp2126_60_S108_L001
 X X X X X Yp2126_65_S113_L001 X X X X X

(111) TABLE-US-00016 TABLE 3 Primers for detecting the presence of nucleic
 acids from *Bacillus anthracis*. Assay SEQ Assay name Type Target species/gene primer
 name sequence (5' -> 3') ID NO CP008853.1_5309 PA *B. anthracis* F Ba-specific-
 3F_UT1 ACGTCAGGTGATTATTGGAC 1 R Ba-specific-3R_UT2
 CAACAATTATATCCGCCATT 2 CP008853.1_5316 PA *B. anthracis* F Ba-specific-5F_UT1
 GAAGATGTACGCTCGATAGG 3 R Ba-specific-5R_UT2 GAAATTCTTTTGGCCATCAC 4
 CP012725.1_3629 PA *B. anthracis* F Ba-specific-6F_UT1 CACAATTGAATGAAAATGCT 5 R
 Ba-specific-6R_UT2 CACGAAACCTGTTTACCTTT 6 CP012725.1_5103 PA *B. anthracis* F
 Ba-specific-8F_UT1 GATATTCGACGAGCTTTCTG 7 R Ba-specific-8R_UT2
 TATTCATCGTCATCCTCCTC 8 CP012725.1_5107 PA *B. anthracis* F Ba-specific-9F_UT1
 TATTGAACGCATTGAATCAG 9 R Ba-specific-9R_UT2 TATTGGTAAGCAAACCGTCT 10
 JSZQ01000034.1_220 PA *B. anthracis* F Ba-specific-11F_UT1

GGTTACGACAAAATGTAGC 11 R Ba-specific-11R_UT2 TAACTTCTGAAGCGAAAACC
12 JSZS01000036.1_5 PA B. *anthracis* F Ba-specific-12F_UT1 GCGAATTTTAGACGACAATC
13 R Ba-specific-12R_UT2 TAACCGTGCTTAATTCGTTT 14 LGCC01000010.1_232 PA B.
anthracis F Ba-specific-14F_UT1 ATTAATAAGGCGACTGGTGA 15 R Ba-specific-14R_UT2
TTACCCATCCAGAATGAGAC 16 LGCC01000048.1_280 PA B. *anthracis* F Ba-specific-
16F_UT1 ACAATTCTTAAAAGGCGACA 17 R Ba-specific-16R_UT2
TGTAGCGTCTCCGATATTTT 18 NN_LOMU01000090.1_49 PA near neighbor F Ba-
specific-20F_UT1 CATGGGGCTTTCTATTATGT 19 species R Ba-specific-20R_UT2
TTCGTTCTTTCATAAGTTTCCT 20 NN_LOQC01000013.1_3 PA near neighbor F Ba-
specific-22F_UT1 TTGGAGTTTGTGTTTGCTTTT 21 species R Ba-specific-22Rv2_
GTAACAATTAATCCACGTCCT 22 UT2 ChimpKiller_9-159 PA B. *cereus* spp. F
ChimpKiller_9F TTATCGTCCATTCTTTCGTC 23 *anthracis* R ChimpKiller_159R
AAACCTAATGAAACGGGATT 24 ChimpKiller_91-320 SV B. *cereus* spp. F
ChimpKiller_91F TATGAAAGGAGCCGTAAAC 25 *anthracis* R ChimpKiller_320R
TGAATATGAAGCGGAAAAC 26 ChimpKiller_481-698 SV B. *cereus* spp. F
ChimpKiller_481F TCGAACATACCTCCATTCT 27 *anthracis* R ChimpKiller_698R
AAAGATAGCTTTGCACTTGG 28 plcR PA Virulence locus plcR F Ba-specific-1F_UT1
TTTTTCGTAAGCATCTTCAA 29 R Ba-specific-1R_UT2 TTTGATGTGAAGGTGAGACA 30
pagA PA Virulence locus pagA F 801F_pagAv3_UT1 GGTTACAGGACGGATTGATA 31 R
1042R_pagAv3_UT2 TCCCACCAATATCAAAGAAC 32 pX01 PA Virulence plasmid F
pX01_113F_UT1 TGAGCCTACCTAGTGATTGG 33 pX01 R pX01-315Rv2_UT2
TTGGATAAATTCCACAAATTCC 34 TC pX02 PA Virulence plasmid F pX02_101F_UT1
CGCCAGCGTATTATATAGGT 35 pX02 R pX02_269R_UT2 GCTAATTCTGGGTTGTGTTT 36
gyrA SNP Drug resistance SNP F gyrA_28Fv2_UT1 TCGGTAAGTATCACCTCA 37 gyrA R
gyrA_182Ry2_UT2 TGCTTCTGTATAACGCATT 38 parC SNP Drug resistance SNP F
parC_1F_UT1 CAGTCGGTAACGTTATTGGT 39 parC R parC_197R_UT2
TAACTCAGATGCAATTGGTG 40 gyrB SNP Drug resistance SNP F gyrB_8F_UT1
ATTGTAGAGGGTGACTCTGC 41 gyrB R gyrB_194R_UT2 TATCAAAATCTCCGCCAAT 42
rpoB SNP Drug resistance SNP F rpoB_29F_UT1 TTCTTCGGAAGTTCTCAGTT 43 rpoB R
rpoB_196R_UT2 CGGACACATACGACCATAG 44 AA_2502 SNP Drug resistance SNP F
AA_2502_UT1 AAGTTTGAGGTGTGGAAATG 45 R AA_2502_UT2
TCGAAATGAGTTCCAATTTT 46 AA_2503 SNP Drug resistance SNP F AA_2503v2_UT1
CAAACTAATAGGGGAGGGTG 47 R AA_2503_UT2 CCGAGAACCTACCTCGTTA 48
Ba_AmesAnc_4669915 SV near neighbor species F Ba&NN32_F
AGGAGATGAGAGTTTTGCAC 49 R Ba&NN32_R ACCCCCATAATTACCATGA 50
Ba_AmesAnc_4001578 SV near neighbor species F Ba&NN33_F
CGTTGCGTAAGTATGTGCTA 51 R Ba&NN33_R AGGTGGCGTAATTAACGTAG 52
Ba_AmesAnc_1069024 SV near neighbor species F Ba&NN37_F
CGAAAAGTTGTCGACCTAAT 53 R Ba&NN37_R ACTGCGTTCACGAAGAATAG 54
Ba_AmesAnc_3668548 SV near neighbor species F Ba&NN38_F
TCTCTTGATTCAACGTTTCC 55 R Ba&NN38_R GATGCAAAACCAATTCATT 56
Ba_AmesAnc_371913 SV near neighbor species F Ba&NN40_F
GTGAAACATCGCTTTTTAGG 57 R Ba&NN40_R TCCGCAATGATATACTTCAA 58
Ba_AmesAnc_999035 SV near neighbor species F Ba&NN41_F
ATACGGTGAAAATGAAGCAG 59 R Ba&NN41_R CGTCTTTGGTAATCGTTCA 60
(112) TABLE-US-00017 TABLE 4 Primers for detecting the presence of nucleic
acids from *Burkholderia*. Assay Target species/ sequence SEQ ID Assay name Type gene
primer name (5' -> 3') NO: TTS1_BPSS1407 PA TTS1 F BpAmpSeq_1_F
TCGTGTCACCGGGAT 61 GGTC R BpAmpSeq_1_R GGCCTTTGCCCGCATA 62 CTCG
LXCC01000141.1_392 PA B. *pseudomallei* F BpAmpSeq_3_F TCGCAWGAAGTGCGT 63

96_39817 TGCCG R BpAmpSeq_3_R GCCCGTTGCGAAGCGA 64 TGAT
LXBY01000087.1_757 PA B. *pseudomallei* F BpAmpSeq_4_F CGCGCTTGCCCAACTA 65
60_76751 CCAG R BpAmpSeq_4_R GCGCAACGGTGCAG 66 ACAAT
LXCD01000002.1_996 PA B. *pseudomallei* F BpAmpSeq_5_F AATCCATGCATGTTCGY 67
52_100245 GCCC R BpAmpSeq_5_R GCGATCGCTCAACGGG 68 CTTC
LXCE01000123.1_342 PA B. *pseudomallei* F BpAmpSeq_6_F TCGCATTGTCAYACGC 69
20_34747 TCCC R BpAmpSeq_6_R AGTGCGCAAACCTTGGC 70 GAGG BpCEN586498 PA
pseudomallei F BpCEN586498_F102 CACCGAAAGATTTTCAG 71 TTCCGCCTCATTCA R
BpCEN586498_R388 GGCCGTCGATGGTTTC 72 GTCGGTTTTTC BpCEN617822 PA
pseudomallei F BpCEN617822_F43 TGCATTGAGCACGGCA 73 CGCAGATTC R
BpCEN617822_R260 GAAAAATTTATCGGAT 74 CGAGCACCATGGTTTG BpCEN972235 PA
pseudomallei F BpCEN972235_F107 ATACGCGGCGCGGCTC 75 ATTTTCG R
BpCEN972235_R305 GCGTCGCGCTCGTCGA 76 TACGGTCA BpCEN70178 PA *pseudomallei*
F BpCEN70178-2F TGCGCAGCGAGTGGTT 77 CAGGTTGTC R BpCEN70178-182R
CGACGATACGGATAC 78 GGCACGGAAGC BpCEN508364 PA *pseudomallei* F UT1-
BpCEN508364_F37 CCGCGCCGGCCGCGAG 79 ACC R UT2-BpCEN508364_R184
CGGGCGTGCCGGACTC 80 CTCGTC LWWC01000187.1_18 PA B. *pseudomallei* F
BpAmpSeq_8_F CCTTTGCGGCAAGCGT 81 *mallei* CGAA R BpAmpSeq_8_R
GAGCCAACGCACATG 82 GACGG LWWB01000125.1_17 PA B. *pseudomallei* F
BpAmpSeq_10_F CCAGTCGGGCGCGGA 83 183_17602 *mallei* AAAAC R BpAmpSeq_10_R
GGCGGCAAAAGCGTC 84 GATGA LXAY01000367.1_0_6 PA B. *pseudomallei* F
BpAmpSeq_11_F GCCGGAACCGTCGAG 85 40 *mallei* R BpAmpSeq_11_R CATTG
TGGATTCGACTGCCTC 86 CGCT LWVY01000190.1_17 PA B. *pseudomallei* F
BpAmpSeq_12_F TCGATATCCGCCGTCT 87 226_17689 *mallei* CGCC R BpAmpSeq_1_R
ATGTGTCGGTGGGCTT 88 CGGT LXAD01000059.1_247 PA B. *pseudomallei* F
BpAmpSeq_13_F GAAAGGCGATGTGCC 89 60_25075 *mallei* GAGCG R BpAmpSeq_13_R
TTCGGAGAAGCGCCA 90 AACGC BpmCEN322640 PA *pseudomallei/mallei* F
BpCEN322640_F2 CGCGGACAGCATCGAT 91 TACGTGAATC R BpCEN32264_R2
CCGCCGAATCCGATGC 92 TCAATTTTC BpmCEN1761486 PA *pseudomallei/mallei* F
BpmCEN1761486_F1 GACCTGCAGCAGGTAT 93 TCGACATTATCGTTC R
BpCEN1761486_R1 AGCTTCGCATACAGCA 94 CTTCCGCCAG BpmCEN1235988 PA
pseudomallei/mallei F BpmCEN1235988_F1 GCGCTGCCCCGTTTCAC 95 CACTGG R
BpmCEN1235988_R1 CGTGACGCCGTCGGGA 96 AAGATCATC BpmCEN1565214 PA
pseudomallei/mallei F BpmCEN1565214_F1 CTGACCGAACGATGGC 97 TGGAGATACATGC
R BpmCEN1565214_R1 CAAATGGGAAGCGAG 98 CTCCCTTCCGA BpmCEN276339-1 PA
pseudomallei/mallei F BpmCEN276339-1_F1 CGGACGCCTGTCGCCC 99 GAAACCTAT R
BpmCEN276339-1_R1 CGCGAGCACGCCGAG 100 CGACAT BpmCEN276339-2 PA
pseudomallei/mallei F BpmCEN276339-2_F1 CGTCGACGCCCCGGGC 101 TTTCTG R
BpmCEN276339-2_R1 CGCCGCGCACCGGTTT 102 CAATC BpmCEN894337 PA
pseudomallei/mallei F BpmCEN894337_F_1 CGAAAATAATTTTCGG 103 CCGGCGCAC R
BpmCEN894337_R1 CGACAGGCATCGGGC 104 GACTACTACCAG BpmCEN1722622 PA
pseudomallei/mallei F UT2-Bpm_CEN1722622-f1 CAACGGGCGAGTTTGC 105 AACGGAATC
R UT1-Bpm_CEN1722622-1-1 GCCGGCTTGGCTTCGT 106 CCTTGTC BpmCEN357268 PA
pseudomallei/mallei F UT1-Bpm_CEN357268-fl CGGCATGCGCGGCCG 107 AATC R UT2-
Bpm_CEN357268-r1 ATCGCGCCCTGCAGCG 108 AGCAC NC_006350_2289827 SV B.
pseudomallei F BpAmpSeq_16_F GCCAGCGCATCCACCA 109 complex SNP ACAT R
BpAmpSeq_16_R AGAGGAAGAAGGGCG 110 AGGCG NC_006350_133027 SV B.
cepacia complex F BpAmpSeq_18_F CGCGCARYTCGTCTGTC 111 SNPs CTCG R
BpAmpSeq_18_R CGAACCTSGTGCMGGT 112 RCAG NC_006350_2248145- SV B.
pseudomallei F BpAmpSeq_19_F CACGTTGCCSGGRAAR 113 2248193 complex SNP TACG

R BpAmpSeq_19_R CCGTCGACAAATCGC 114 GCTS NC_006350_988041- SV B.
pseudomallei F BpAmpSeq_20_F CAGAACGCGCTRTYCC 115 988089 complex SNP ACG R
BpAmpSeq_20_R TGCCGCGTGATCCATT 116 GCAT Bm_11589 PA B. *mallei* F
BpAmpSeq_21_F AGGGGGTGGTTTCCTG 117 AGTGGCGTGAC R BpAmpSeq_21_R
AGCGGTGTCGACGGGT 118 GGAAAGGATG Bm_11767 PA B. *mallei* F BpAmpSeq_22_F
ACGGGCGCTTCACGAT 119 CTCGGTGTTT R BpAmpSeq_22_R GCGCGGCAGTTTCGATC
120 AGGCATTTG Bm_11589 PA B. *mallei* F Bm-11589-f1_UT1 GACGGCGGGCTTTGGG
121 GAGTCC R Bm-11589-r1_UT2 GCTCGCGGGCAGCGGT 122 GTCG Bm_11767 PA B.
mallei F Bm-11767-f1_UT1 GACGGCCCCGGGCGG 123 CTTTAC R Bin-11767-r1_UT2
CGCGGCAGTTTCGATCA 124 GGCATTTGAG Bm_11767 PA B. *mallei* F Bm-11767-f1_UT1
GACGGCCCCGGGCGG 125 CTTTAC R Bm-11767-r2_UT2 CGAGGGGCGAAATTC 126
CCCTTATAGATCAGTT G K9penA378-529 SNP penA F BpAmpSeq_26_F
CGGTCCGACAAATTC 127 GCACGCACTC R BpAmpSeq_26_R AGCGAGCGGCGCAAC
128 GGAGAATGATT K9penA575-761 SNP penA F BpAmpSeq_27_F GCTGCGCGGCCAAGC
129 GAAAAACG R BpAmpSeq_27_R CGCGAGGACCGCAGC 130 GCAAAGC K9penA949-
1172 SNP penA F BpAmpSeq_28_F GGCCGCAGACCGTCAC 131 CGCGTATG R
BpAmpSeq_28_R GTCGCCCCGTCTTGTTG 132 CCGAGCATC penA_-78promoter SNP penA F
K9penA281fUT1 GCCCGTCAATCCGATG 133 CMGTATCTGG R K9penA565rUT2
GCGCCGATCARTGGGG 134 TGGAAATG penA_C69Y_S72F SNP penA F K9penA696fUT1
CATCGCGGCGACGAG 135 CGTTTCC R K9penA848rUT2 CTCGGTGATCGGCGAA 136
TAGCGGATGAGA penA_P167S SNP penA F K9penA881fUT1 GCTGTGCGCGGCGACG 137
CTTCAGTA R K9penA1258rUT2 CCGATGTCGTTCGCCG 138 TTCCGTAGTC PBP3-170f-
505r PA penA F PBP3-170f3UT1 ATCCGCCGTCCCGCCC 139 AGCAATAG R PBP3-505r3UT2
GGGTTCGCCCAGATTT 140 CGTAGGTGGTGAG pbp3-1 PA pbp3 F K9pbp336fUT1
TCGCCGTTTCACGCCC 141 CGCAAC R K9PBP3331rUT2 GCGCCGAACGCGAGG 142
AACACGA pbp3-2 PA pbp3 F K9pbp31292fUT1 GCTCGGAAGCTCGCG 143 CTGAACC R
K9PBP31527rUT2 GGATCGTGCCGTCGCC 144 CGCATAC V15G_R20 SV folM *pteredine* F
1026pter371fU ACAAGCCCGGYGTCGTCTG 145 reductase AGATGGTGAC R
1026pter636rUT2 CGCGTCGGCCGAAYG 146 GTCGTAGT bpeT HTH SV bpeT HTH
region F bpeT_-76fUT1 AATCGTCGGCTGCGTC 147 GCCTTCA R bpeT_596rUT2
CGGGTAGCGTGAGTG 148 GAATTCGCAGAG bpeT substrate SV bpeT substrate F
bpeT_695fUT1 CCTCGAAGGCTTCGGG 149 binding binding region CTGATCCAG R
bpeT_1014rUT2 GACTAACCGCTTACGC 150 CACCCACTCGTTC bpeS HTH SV bpeS
HTH region F bpeS_-83fUT1 AAAGCGAATAGTCGC 151 GAAGCGGCTTGA R
bpeS_230rUT2 GCGATCTCGGTGATGA 152 TCTTGATGCAGTG bpeS substrate SV
bpeS substrate F bpeS_648fUT1 AACGGCGGCGTGACC 153 binding binding region
GTCAACG R bpeS_977rUT2 CGCTACGCGGCCACCT 154 GCCC bimA PA bimA F
Bpvir_bimA_407F CGGAGCTTCAGAACA 155 ACCCGCGTGTAAC R Bpvir_bimA_654R
CCTTCGGACCTTTTCC 156 CGCAACTGGC cheD PA cheD F Bpvir_cheD_29F
AATTCGGCCGGCAGGC 157 GGTACG R Bpvir_cheD_297R CGCGCGCAGCCGGCAT 158
TTG fhaB1long PA fhaB1long F Bpvir_fhaB1long_8410F CCCTTCGGTCCCCACC 159
AGAAAAATTCTG R Bpvir_fhaB1long_8599R AGCCGTACAGGCCAAT 160
GCAGCCATCTATG fhaB1short PA fhaB1short F Bpvir_fhaB1short_63F
GCGCCGCGTGTTCTGTG 161 ACCTTGTC R Bpvir_fhaB1short_316R CGCTGATCGGCGCATC
162 GGACAC fhaB2 PA fhaB2 F Bpvir_fhaB2_1812F ATCGTGATATCGCCGG 163
TTCCTGGTTGTG R Bpvir_fhaB2_2100R CACGTTTGGCGGCAGT 164 GCAAGGTGTAG
fhaB3 PA fhaB3 F Bpvir_fhaB3_3966F TCTGCTGATCGGCCTT 165 CGCCAGATAYAC R
Bpvir_fhaB3_4324R GCGGATGAACAATTTC 166 CTGTCGAGCGACTATT AC LPSA PA
LPSA F Bpvir_LPSA_1087F GCAGGGCGCCTTGATA 167 TCCGCTATGAG R
Bpvir_LPSA_1407R CGGCGCAAGGTTCTCC 168 TGCCACATC LPSb1 PA LPSb1 F

Bpvir_LP_Sb1_65F GTGTGATGTGATCGACKGCT 169 CCTCCCTGAG R Bpvir_LPSb1_256R
CAAGCCGCTGATACCC 170 GTGTCGCTG LPSb2 PA LPSb2 F Bpvir_LPSb2_88F
GCGCTTCTCGGTGGGT 171 ACGAAAAACAGC R Bpvir_LPSb2_400R
CGAGTCGGCCAAGATC 172 ATTCAGGACCAG wcbj PA wcbj F Bpvir_wcbj_252F
CACCTTGACACTGATC 173 CGCGGCGTAG R Bpvir_wcbj_508R CTTCTTCGCACAACC
174 GAGCAAATACTGAGT AAATC ylf PA ylf F Bpvir_ylf_865F GATCTTGCGACCGATG 175
CTCAGCGTGTG R Bpvir_ylf_1153R TGGCGCGGGCCAAGG 176 ATATCAGTTC
thai_small_15666 PA *thailandensis* (small F thai_small_15666_3F_UT1
GCCTTACGCCTTCGGG 177 clade) ATCG R thai_small_15666_342R_UT2
GAATGCGCTCACCCGA 178 TGCT thai_small_28301 PA *thailandensis* (small F
thai_small_28301_11F_UT1 AGCAAGCCATCCGCGT 179 clade) CATC R
thai_small_28301_297R_UT2 CAGGATGCCACCGTTG 180 GTGA thai_large_48054 PA
thailandensis (large F thai_large_48054_286F GCCACAGGCATGGTG 181 clade) AGCAA R
thai_large_48054_534R CGGCATTCCCTCAATC 182 ACGAA thai_all_110625 PA *thailandensis*
F thai_all_110625_212F CTGCGTCCCAAACCGA 183 CGA R thai_all_110625_512R
CCGTTCGATGCCACGAA 184 TGAA humpty_7099 PA *humptydooensis* F humpty_7099_510F
CCCCAAAAATCCCGCT 185 CTGG R humpty_7099_762R CGGCACAAAGCCGGT 186
GAAAG humpty_45647 PA *humptydooensis* F HUMPTY_45647_173F_UT1
TGCCGTTTCAGTTGGGC 187 CTTT R HUMPTY_45647_390R_UT2 TGCCGCTTCCAACCTGC
188 TTCA humpty_7093 PA *humptydooensis* F HUMPTY_7093_48F_UT1
GGGCGGGCCAATCTTT 189 TCTG R HUMPTY_7093_381R_UT2 TCCGCGATGTGACCAA
190 ACGA humpty_38764 PA *humptydooensis* F humpty_38764_44F TCGGAGATTCCGACGG
191 ACCA R humpty_38764_390R CCGCATATCGCCCTGA 192 CACA okla_24632 PA
oklahomensis F OK_24632_724F_UT1 GGCACCGACGTGCAA 193 AAAGC R
OK_24632_996R_UT2 GGCCGATCTCGGCACT 194 ACGA okla_5812 PA *oklahomensis* F
okla_5812_382F GCGGGGTACGGGCTA 195 ACCAA R okla_5812_700R
TCCGTACGCTCGCCAC 196 AACA okla_3784 PA *oklahomensis* F okla_3784_454F
GCAAAGGCGCCAGGA 197 AACAA R okla_3784_742R ACCGCCCCGATTGACC 198 AAGT
okla_like_18345 PA *oklahomensis*-like F OK_like_18345_56F_UT1 TCCAGGCGGTTCTCCG
199 ATTG R OK_like_18345_372R_UT2 GTTGCCGATGTGAGG 200 CACA okla_like_18342
PA *oklahomensis*-like F OK_like_18342_361F_UT1 TCTTCGGCGAGCGTCT 201 ACGG R
OK_like_18342_685R_UT2 CGCGTCGGACGAGTGT 202 CGTA MSMB175_14005 PA
MSMB175 group F MSMB175_14005_422F GGCTCACACGGCTGGG 203 TCAT R
MSMB175_14005_746R ACGGCGTTTTTGGACCA 204 CGAG MSMB175_1868 PA
MSMB175 group F MSMB175_1868_197F_UT1 CCGCCTACTGGTGGA 205 GGTG R
MSMB175_1868_480R_UT2 GCCAGTCCCGGGAAG 206 GAGTG MSMB175_8900 PA
MSMB175 group F MSMB175_8900_29F_UT1 GCTCATCCTGCCAGGC 207 CAGT R
MSMB175_8900_345R_UT2 GATACCCACCGCCGGA 208 ACCT MSMB175_9798 PA
MSMB175 group F MSMB175_9798_517F_UT1 AGCGGCGGATTATGG 209 GCACT R
MSMB175_9798_782R_UT2 ACGCTGGGGCTGTTTT 210 GCAG MSMB264_34074 PA
MSMB264 group F MSMB264_34074_554F CGCCCTTCGAGCTTGC 211 TTCC R
MSMB264_34074_778R CCGCAACAGGTGGCTT 212 CTGAC MSMB264_4163 PA
MSMB264 group F MSMB264_4163_252F CGTTGCCCCCGCCAC 213 GTAG R
MSMB264_4163_596R CCGTGTGGCGCGTCCT 214 CCAT vietnam_61292 PA *vietnamiensis* F
vietnam_61292_183F TGGGCTCATCCTCGCA 215 AAGC R vietnam_61292_527R
ACGCGCTCGGTGGAA 216 AACAG vietnam_98057 PA *vietnamiensis* F vietnam_98057_111F
TCACACCATGGGCTCC 217 GAGA R vietnam_98057_460R CGGGCGGGTAGACGA 218
GTTCC vietnam_226017 PA *vietnamiensis* F vietnam_226017_33F_UT1
ACCACGAGTGTGTGCG 219 GCATT R vietnam_226017_285R_UT2
GCGCTCGATGGTTCCC 220 GAAG ubon_small_102920 PA *ubonensis* (small F

ubon_small_102920_511F CTTGCCTTCCAGGCGC 221 clade) ACAT R
 ubon_small_102920_802R TGCCAAGCGGAAGCTC 222 CTTG ubon_small_111449 PA
ubonensis (small F ubon_small_111449_167F_UT GCCGTGTCCGCATGAT 223 clade) 1 CCTC
 R ubon_small_111449_431R_U CGCTCCAGTGCGTTGT 224 T2 CGAG ubon_large_1438777
 PA *ubonensis* (large F ubon_large_41F_1438777_BH CACTGTTCGCATCGGT 225 clade) _RP
 ATT C R ubon_large_240R_1438777_B CTYGCCGTGTCCGTCA 226 H_RP CGACAAG
 ubon_all_1328624 PA *ubonensis* F ubon_all_1328624_220F GGCGCCTTCTGGTGGT 227 CCTT
 R ubon_all_1328624_563R TGGCTTTGCGACCAGT 228 CGTG *cepacia*_1208120 PA *cepacia*-
 complex F *cepacia*_1208120_1F ATGGCAARGATTCTKG 229 TRG R *cepacia*_1208120_311R
 TTCACGATCCAGCCCT 230 T

(113) TABLE-US-00018 TABLE 5 Primers for detecting the presence of nucleic
 acids from *Yersinia*. SEQ Assay Target species/ ID Assay name Type gene primer name
 sequence (5'→3') NO: Ypestis_ PA Y. *pestis* F Yp&NN1_F AACAAGCTAAAACCGAACAA
 231 LPQY01000176.1_7 R Yp&NN1_R ATAGCCTCAACTGCTTTTTTG 232
 AGJT01000065.1_0_338 PA Y. *pestis* F Yp&NN11_F CAGTACCGACAAAACCTTC 233 R
 Yp&NN11_R TTTACTACTCTGAAAACGAG 234 FAUR01000053.1_96407_ PA Y. *pestis* F
 Yp&NN12_F GCACTACAAATTAAATCCC 235 96884 R Yp&NN12_R
 GTCGATTATCAACCTCTATG 236 Wagner_Yp_pla_Forward PA Y. *pestis* F Yp&NN2_F
 GAAAGGAGTGCGGGTAATAGG 237 TT R Yp&NN2_R GGCCTGCAAGTCCAATATATGG
 238 YpPGM_8-158 PA Virulence locus F YpPGM_8F TTAATATCCCGGCACTCATA 239 PGM
 R YpPGM_158R TCCTTAACTGAATAAGTGCTCA 240 YpPGM_31-205 PA Virulence locus F
 YpPGM_31Fv2 TTTAATGAACGGTGCCTAG 241 PGM R YpPGM_205Rv2
 GTCTGCGTTTCTCCAGTAT 242 Yp-p1202_42780-43194 PA Drug Resistance F Yp-p1202_
 TCTGGCCTGCTAAATAAAAACG 243 plasmid p1P1202 42780F-UT1 AACC R Yp-p1202_
 CAGGCCTCAGCATTTTATTATG 244 43194R-UT2 GTGAT Yp-p1202_126386-126750 PA
 Drug Resistance F Yp-p1202_ GGGGCGGATACCTTCACCTATG 245 plasmid p1P1202
 126386F-UT1 R Yp-p1202_ CTGGGGTTCAGTCTGGACGAGA 246 126750R-UT2 T Yp-
 p1202_156402-156711 PA Drug Resistance F Yp-p1202_ ACCATCCGGCGCTAAATCGTC 247
 plasmid p1P1202 156402F2-UT1 R Yp-p1202_ GAAATGCGCCTGGTAAGCAGA 248
 156711R-UT2 GT YpCO92_NC_003143_113190 SV near neighbor F Yp&NN4_F
 ACTCGGGATACTCCATACTG 249 species R Yp&NN4_R CGAAAGCAGTGGTCAATC 250
 YpCO92_NC_003143_161621 SV near neighbor F Yp&NN5_F CATGCGCTTTACGTTATATG
 251 species R Yp&NN5_R GCGTTCTGCACTCTGTCT 252 YpCO92_NC_003143_152213 SV
 near neighbor F Yp&NN6_F AGCGACTTCCGTGATAAAG 253 species R Yp&NN6_R
 ACTCAGGATACCGTGTGGT 254 YpCO92_NC_003143_129539 SV near neighbor F
 Yp&NN7_F TTCACGATAATCCCCTAATG 255 species R Yp&NN7_R
 TTCTGTGCTCTGGCTGATA 256 YpCO92_NC_003143_91203 SV near neighbor F
 Yp&NN8_F ATTATCTGTGCCCCTTCTTT 257 species R Yp&NN8_R
 GGAGTGGATGCCACTAAAC 258 YpCO92_NC_003143_121812 SV near neighbor F
 Yp&NN9_F CCTCACACAACAATTCACTG 259 species R Yp&NN9_R
 TTTTCCGACAAATTTAAGG 260 Yp_AL590842.1_RX_SNP SV near neighbor F
 Yp&NN10_F AGCATGAAGGTTGCTAAAAG 261 species R Yp&NN10_R
 GGTGACTTCAAAACCGTTAG 262 Yentero_FR729477.2_1623 PA Y. F Yp&NN3_F
 GATGCTTCTGCTATCAGSTT 263 *enterocoliticus* R Yp&NN3_R
 GTGTGRCTTTGAASTCTTGT 264

(114) TABLE-US-00019 TABLE 6 Primers for detecting the presence of nucleic
 acids from *Francisella*. Target SEQ Assay species/ primer sequence ID Assay name Type
 gene name (5'→3') NO: Ftularensis_ PA F. F Ft& GAAGTGGCTC 265 CP000915.1_ *tularensis*
 NN2_F ATGTTAGAGG 1782 R Ft& AGCGAGCCTA 266 NN2_R TATGTAACCA Ftularensis_
 PA F. F Ft& TTTAATGTCC 267 CP000915.1_ *tularensis* NN3_F GTCAACCTCT 731 R Ft&

ACGAGTTTGT 268 NN3_R GAGTCGCTAT Ft_dup_ PA F. F Ft& TGTTACGTAC 269
 CP000915.1_ *tularensis* NN8_F AGGCTGTCAA 197 R Ft& ATCATATCCC 270 NN8_R
 GTAGCACAAG FtA1 SNP FtA1 Clade F 9F_ CATAACCCAT 271 FtA1_ CGCAATATCT UT1
 R 246R_ AAATTATCTG 272 FtA1_ TAGCGGCAAA UT2 FtA2 SNP FtA2 Clade F 34F_
 GTGTCCAACG 273 FtA2_ AAACCATAAT UT1 R 169R_ TTTGGTTGAT 274 FtA2_
 TCTGTCAGTG UT2 FtB SNP FtB Clade F 28F_ AAGCTTAACT 275 FtB_ GGTGATTGGA
 UT1 R 173R_ CGCCTAACAT 276 FtB_ CTTATCTGCT UT2 FtA SNP FtA Clade F 14F_
 GGGTGATGCA 277 FtA_ GTAGAGAAAA UT1 R 207R_ TACCAGATGA 278 FtA_
 ACGAATAGCC UT2 FtLVS_ SV near F Ft& ATCAAGCTCA 279 AM233362_ neighbor NN9_F
 TCTTCAAAGC 1646546 species R Ft& AACCATGTTC 280 NN9_R AGATCCAAAA FtLVS_
 SV near F Ft& TACCTCTGCC 281 AM233362_ neighbor NN10_F AAAAATTCAT 1643765
 species R Ft& GGCATACTCA 282 NN10_R AGGTAGTGGT FtLVS_ SV near F Ft&
 TCTTTGGTAG 283 AM233362_ neighbor NN11_F CTTGCTGACT 1562618 species R Ft&
 CAGACGACAC 284 NN11_R TTGGCTTATT Ftnovicida_ PA F. F Ft& GG TAGGATAA 285
 CP009607.1 *tularensis* NN1_F CTACCAAG spp. R Ft& GTCATGAGTT 286 *Novicida* NN1_R
 TTACCAATAC TC Fphilom_ PA F. F Ft& CTTATGCAGC 287 CP009444.1_ *philomiragia*
 NN6_F AAGAGGAACT 569 R Ft& ATACACCGGG 288 NN6_R ATAGGTTTCT Fphilom_ PA
 F. F Ft& CTGATGGAAG 289 CP009444.1_ *philomiragia* NN7_F AGAGTTTCGAG 285 R Ft&
 GTAGATATAA 290 NN7_ TCAGCGCCAC Rv2 Fnoatunensis_ PA F. F Ft& CGGTAAGAAT 291
 CP003402.1_ *noatunensis* NN4_F ACGACCAGAG 1749 R Ft& AGAGGATTTC 292 NN4_R
 TTCCTCCTTG Fnoatunensis_ PA F. F Ft& AATTCTACAA 293 CP003402.1_ *noatunensis*
 NN5_F GCACCTGGAA 424 R Ft& TCCTATTAAA 294 NN5_R AGCGCCATAG

(115) TABLE-US-00020 TABLE 7 Primers for Sequence Control. For- Re- Target ward
 Forward SEQ verse Reverse SEQ Assay species/ primer sequence ID primer sequence ID name
 gene name (5'→3') NO name 5'→3') NO IPSC- se- UT1- GGGCGGAC 295 UT2- GCCGGGAT
 298 1 quencing IPSC- GAAAACCC IPSC- GCCTTACC control f1 TTGAGCAC r1 TAGACGCA
 AG ATGA IPSC- se- UT1- GCTCGGGC 296 UT2- GCCGGGAT 299 1 quencing IPSC-
 GGACGAAA IPSC- GCCTTACC control f1v2 ACCCTTGA r1 TAGACGCA ATGA IPSC- se-
 UT1- GCGGCAGC 297 UT2- CGAGTTCC 300 2 quencing IPSC- CGTTGAGG IPSC-
 GTCCGGTT control f2 CAAAAGTG r2 AAGCGTGA ATAC CAGTC Forward sequence w/UT:
 ACCCAACTGAATGGAGC (SEQ ID NO: 301) at 5' of the forward sequence, e.g., UT1-IPSC-
 f1: ACCCAACTGAATGGAGCGGGCGGACGAAAACCCTTGAGCACAG (SEQ ID NO: 302)
 Reverse sequence w/UT: ACGCACTTGACTTGTCTTC (SEQ ID NO: 303) at 5' of the reverse
 sequence, e.g., UT2-IPSC-r1:

ACGCACTTGACTTGTCTTCGCCGGGATGCCTTACCTAGACGCAATGA (SEQ ID NO: 304)

(116) TABLE-US-00021 TABLE 8 Preparation of Primer Mixture for detecting the presence of
 nucleic acids from *Burkholderia*. uM in mix Volume starting Desired stock needed primer
 Recalculated final (final in Mix conc. to final conc. of Start uM uM × stock add in mix primer in
 Assay name Primer Target (uM) in rxn. 5.56) (uL) stock a single rxn. TTS1_BPSS1407
 BpAmpSeq_1_F TTS1 100 0.10 0.56 0.03 5.0 0.10 TTS1_BPSS1407 BpAmpSeq_1_R TTS1 100
 0.10 0.56 0.03 5.0 0.10 LXCC01000141.1_39296_39817 BpAmpSeq_3_F *pseudomallei* 100 0.20
 1.11 0.05 9.9 0.20 LXCC01000141.1_39296_39817 BpAmpSeq_3_R *pseudomallei* 100 0.20 1.11
 0.05 9.9 0.20 LXBY01000087.1_75760_76751 BpAmpSeq_4_F *pseudomallei* 100 0.20 1.11 0.05
 9.9 0.20 LXBY01000087.1_75760_76751 BpAmpSeq_4_R *pseudomallei* 100 0.20 1.11 0.05 9.9
 0.20 LXCD01000002.1_99652_100245 BpAmpSeq_5_F *pseudomallei* 100 0.40 2.22 0.10 19.8
 0.40 LXCD01000002.1_99652_100245 BpAmpSeq_5_R *pseudomallei* 100 0.40 2.22 0.10 19.8
 0.40 LXCE01000123.1_34220_34747 BpAmpSeq_6_F *pseudomallei* 100 0.40 2.22 0.10 19.8 0.40
 LXCE01000123.1_34220_34747 BpAmpSeq_6_R *pseudomallei* 100 0.40 2.22 0.10 19.8 0.40
 LWWC01000187.1_18 BpAmpSeq_8_F *pseudomallei mallei* 100 0.30 1.67 0.08 14.9 0.30
 LWWC01000187.1_18 BpAmpSeq_8_R *pseudomallei mallei* 100 0.30 1.67 0.08 14.9 0.30

LWWB01000125.1_17183_17602 BpAmpSeq_10_F *pseudomallei mallei* 100 0.20 1.11 0.05 9.9 0.20
 0.20 LWWB01000125.1_17183_17602 BpAmpSeq_10_R *pseudomallei mallei* 100 0.20 1.11 0.05 9.9 0.20
 LXAY1000367.1_0_640 BpAmpSeq_11_F *pseudomallei mallei* 100 0.10 0.56 0.03 5.0 0.10
 LXAY1000367.1_0_640 BpAmpSeq_11_R *pseudomallei mallei* 100 0.10 0.56 0.03 5.0 0.10
 LWVY01000190.1_17226_17689 BpAmpSeq_12_F *pseudomallei mallei* 100 0.40 2.22 0.10 19.8 0.40
 LWVY01000190.1_17226_17689 BpAmpSeq_12_R *pseudomallei mallei* 100 0.40 2.22 0.10 19.8 0.40
 LXAD01000059.1_24760_25075 BpAmpSeq_13_F *pseudomallei mallei* 100 0.30 1.67 0.08 14.9 0.30
 LXAD01000059.1_24760_25075 BpAmpSeq_13_R *pseudomallei mallei* 100 0.30 1.67 0.08 14.9 0.30
 NC_006350_2289827 BpAmpSeq_16_F *pseudomallei* complex 100 0.40 2.22 0.10 19.8 0.40
 SNP NC_006350_2289827 BpAmpSeq_16_R *pseudomallei* complex 100 0.40 2.22 0.10 19.8 0.40
 SNP NC_006350_133027 BpAmpSeq_18_F *cepacia* complex 100 0.20 1.11 0.05 9.9 0.20
 SNPs NC_006350_133027 BpAmpSeq_18_R *cepacia* complex 100 0.20 1.11 0.05 9.9 0.20
 SNPs NC_006350_2248145-2248193 BpAmpSeq_19_F Bpc MSS 100 0.40 2.22 0.10 19.8 0.40
 NC_006350_2248145-2248193 BpAmpSeq_19_R Bpc MSS 100 0.40 2.22 0.10 19.8 0.40
 NC_006350_988041-988089 BpAmpSeq_20_F Bpc MSS 100 0.20 1.11 0.05 9.9 0.20
 NC_006350_988041-988089 BpAmpSeq_20_R Bpc MSS 100 0.20 1.11 0.05 9.9 0.20
 Bm_11589 BpAmpSeq_21_F *mallei* 100 0.40 2.22 0.10 19.8 0.40
 Bm_11589 BpAmpSeq_21_R *mallei* 100 0.40 2.22 0.10 19.8 0.40
 Bm_11767 BpAmpSeq_22_F *mallei* 100 0.40 2.22 0.10 19.8 0.40
 Bm_11767 BpAmpSeq_22_R *mallei* 100 0.40 2.22 0.10 19.8 0.40
 PBP3-170-505 BpAmpSeq_24_F pbp3 100 0.20 1.11 0.05 9.9 0.20
 PBP3-170-505 BpAmpSeq_24_R pbp3 100 0.20 1.11 0.05 9.9 0.20
 K9penA378-529 BpAmpSeq_26_F penA 100 0.10 0.56 0.03 5.0 0.10
 K9penA378-529 BpAmpSeq_26_R penA 100 0.10 0.56 0.03 5.0 0.10
 K9penA575-761 BpAmpSeq_27_F penA 100 0.10 0.56 0.03 5.0 0.10
 K9penA575-761 BpAmpSeq_27_R penA 100 0.10 0.56 0.03 5.0 0.10
 K9penA949-1172 BpAmpSeq_28_F penA 100 0.10 0.56 0.03 5.0 0.10
 K9penA949-1172 BpAmpSeq_28_R penA 100 0.10 0.56 0.03 5.0 0.10
 IPSC IPSC-f1v2 IPSC 20 0.05 0.28 0.06 12.4 0.05
 IPSC IPSC-r1 IPSC 20 0.05 0.28 0.06 12.4 0.05
 Volume of primer mix in a single rxn: 4.5 uL. Desired volume of primer mix stock: 891 uL.

(117) TABLE-US-00022 TABLE 10 Preparation of Primer Mixture for detecting the presence of nucleic acids from SOP for *Bacillus*, *Yersinia*, and *Francisella* UT-AmpSeq PCR and Bead Cleanup

Desired stock	Primer	Starting final conc.	a single final (final in Mix mix primer conc. Of primer Start rxn uM in uM × stock stock to add in mix in a single Primer name Assay name (uM) (uL) rxn.	5.56) (uL) (uL) stock rxn.	Ba-specific-1F_UT1	plcR	100	4.5	0.1	0.56	0.03	891	5	0	Ba-specific-1R_UT2	plcR	100	0.1	0.56	0.03	5	0	Ba-specific-3F_UT1	CP008853.1_5309	500	0.2	1.11	0.01	2	0	Ba-specific-3R_UT2	CP008853.1_5309	500	0.2	1.11	0.01	2	0	Ba-specific-5F_UT1	CP008853.1_5316	500	0.2	1.11	0.01	2	0	Ba-specific-5R_UT2	CP008853.1_5316	500	0.2	1.11	0.01	2	0	Ba-specific-6F_UT1	CP012725.1_3629	500	1.6	8.9	0.08	15.9	0	Ba-specific-6R_UT2	CP012725.1_3629	500	1.6	8.9	0.08	15.9	0	Ba-specific-8F_UT1	CP012725.1_5103	100	0.1	0.56	0.03	5	0	Ba-specific-8R_UT2	CP012725.1_5103	100	0.1	0.56	0.03	5	0	Ba-specific-9F_UT1	CP012725.1_5107	100	0.1	0.56	0.03	5	0	Ba-specific-9R_UT2	CP012725.1_5107	100	0.1	0.56	0.03	5	0	Ba-specific-11F_UT1	JSZQ01000034.1_220	100	0.1	0.56	0.03	5	0	Ba-specific-11R_UT2	JSZQ01000034.1_220	100	0.1	0.56	0.03	5	0	Ba-specific-12F_UT1	JSZS01000036.15	100	0.1	0.56	0.03	5	0	Ba-specific-12R_UT2	JSZS01000036.15	100	0.1	0.56	0.03	5	0	Ba-specific-14F_UT1	LGCC01000010.1_232	500	0.4	2.22	0.02	4	0	Ba-specific-14R_UT2	LGCC01000010.1_232	500	0.4	2.22	0.02	4	0	Ba-specific-16F_UT1	LGCC01000048.1_280	100	0.1	0.56	0.03	5	0	Ba-specific-16R_UT2	LGCC01000048.1_280	100	0.1	0.56	0.03	5	0	Ba-specific-20F_UT1	NN_LOMU01000090.1_49	500	1	5.56	0.05	9.9	0	Ba-specific-20R_UT2	NN_LOMU01000090.1_49	500	1	5.56	0.05	9.9	0	Ba-specific-22F_UT1	NN_LOQC01000013.1_3	500	0.2	1.11	0.01	2	0	Ba-specific-22Rv2_UT2	NN_LOQC01000013.1_3	500	0.2	1.11	0.01	2	0	pX01_113F_UT1	pX01	500	0.8	4.45	0.04	7.9	0	pX01-315Rv2_UT2	pX01	500	0.8	4.45	0.04	7.9	0
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pX02_101F_UT1 pX02100 0.1 0.56 0.03 5 0 pX02_269R_UT2 pX02 100 0.1 0.56 0.03 5 0
gyrA_28Fv2_UT1 gyrA 50 0.05 0.28 0.03 5 0 gyrA_182Rv2_UT2 gyrA 50 0.05 0.28 0.03 5 0
parC_1F_UT1 parC 100 0.05 0.28 0.01 2.5 0 parC_197R_UT2 parC 100 0.05 0.28 0.01 2.5 0
gyrB_8F_UT1 gyrB 100 0.1 0.56 0.03 5 0 gyrB_194R_UT2 gyrB 100 0.1 0.56 0.03 5 0
801F_pagAv3_UT1 pagAv3 100 0.1 0.56 0.03 5 0 1042R_pagAv3_UT2 pagAv3 100 0.1 0.56 0.03
5 0 rpoB_29F_UT1 rpoB 50 0.05 0.28 0.03 5 0 rpoB_196R_UT2 rpoB 50 0.05 0.28 0.03 5 0
AA_2502_UT1 AA_2502 500 0.8 4.45 0.04 7.9 0 AA_2502_UT2 AA_2502 500 0.8 4.45 0.04 7.9
0 AA_2503v2_UT1 AA_2503 500 0.8 4.45 0.04 7.9 0 AA_2503_UT2 AA_2503 500 0.8 4.45 0.04
7.9 0 Ba&NN32_F Ba_AmesAnc_4669915 100 0.1 0.56 0.03 5 0 Ba&NN32_R
Ba_AmesAnc_4669915 100 0.1 0.56 0.03 5 0 Ba&NN33_F Ba_AmesAnc_4001578 100 0.05 0.28
0.01 2.5 0 Ba&NN33_R Ba_AmesAnc_4001578 100 0.05 0.28 0.01 2.5 0 Ba&NN37_F
Ba_AmesAnc_1069024 500 0.2 1.11 0.01 2 0 Ba&NN37_R Ba_AmesAnc_1069024 500 0.2 1.11
0.01 2 0 Ba&NN38_F Ba_AmesAnc_3668548 500 0.2 1.11 0.01 2 0 Ba&NN38_R
Ba_AmesAnc_3668548 500 0.2 1.11 0.01 2 0 Ba&NN40_F Ba_AmesAnc_371913 500 0.2 1.11
0.01 2 0 Ba&NN40_R Ba_AmesAnc_371913 500 0.2 1.11 0.01 2 0 Ba&NN41_F
Ba_AmesAnc_999035 100 0.05 0.28 0.01 2.5 0 Ba&NN41_R Ba_AmesAnc_999035 100 0.05
0.28 0.01 2.5 0 ChimpKiller_9F ChimpKiller_9-159 100 0.1 0.56 0.03 5 0 ChimpKiller_159R
ChimpKiller_9-159 100 0.1 0.56 0.03 5 0 ChimpKiller_91F ChimpKiller_91-320 500 0.8 4.45 0.04
7.9 0 ChimpKiller_320R ChimpKiller_91-320 500 0.8 4.45 0.04 7.9 0 ChimpKiller_481F
ChimpKiller_481-698 500 0.8 4.45 0.04 7.9 0 ChimpKiller_698R ChimpKiller_481-698 500 0.8
4.45 0.04 7.9 0 Yp&NN1_F Ypestis_LPQY01000176.1_7 500 0.2 1.11 0.01 2 0 Yp&NN1_R
Ypestis_LPQY01000176.1_7 500 0.2 1.11 0.01 2 0 Yp&NN2_F Wagner_Yp_pla_Forward 100 0.1
0.56 0.03 5 0 Yp&NN2_R Wagner_Yp_pla_Forward 100 0.1 0.56 0.03 5 0 Yp&NN3_F
Yentero_FR729477.2_1623 500 0.2 1.11 0.01 2 0 Yp&NN3_R Yentero_FR729477.2_1623 500 0.2
1.11 0.01 2 0 Yp&NN4_F YpCO92_NC_003143_113190 500 0.4 2.22 0.02 4 0 Yp&NN4_R
YpCO92_NC_003143_113190 500 0.4 2.22 0.02 4 0 Yp&NN5_F YpCO92_NC_003143_161621
100 0.05 0.28 0.01 2.5 0 Yp&NN5_R YpCO92_NC_003143_161621 100 0.05 0.28 0.01 2.5 0
Yp&NN6_F YpCO92_NC_003143_152213 100 0.05 0.28 0.01 2.5 0 Yp&NN6_R
YpCO92_NC_003143_152213 100 0.05 0.28 0.01 2.5 0 Yp&NN7_F
YpCO92_NC_003143_129539 100 0.1 0.56 0.03 5 0 Yp&NN7_R YpCO92_NC_003143_129539
100 0.1 0.56 0.03 5 0 Yp&NN8_F YpCO92_NC_003143_91203 100 0.05 0.28 0.01 2.5 0
Yp&NN8_R YpCO92_NC_003143_91203 100 0.05 0.28 0.01 2.5 0 Yp&NN9_F
YpCO92_NC_003143_121812 100 0.1 0.56 0.03 5 0 Yp&NN9_R YpCO92_NC_003143_121812
100 0.1 0.56 0.03 5 0 Yp&NN10_F Yp_AL590842.1_RX_SNP 50 0.05 0.28 0.03 5 0 Yp&NN10_R
Yp_AL590842.1_RX_SNP 50 0.05 0.28 0.03 5 0 Yp&NN11_F AGJT01000065.1_0_338 100 0.1
0.56 0.03 5 0 Yp&NN11_R AGJT01000065.1_0_338 100 0.1 0.56 0.03 5 0 Yp&NN12_F
FAUR01000053.1_96407_96884 100 0.1 0.56 0.03 5 0 Yp&NN12_R
FAUR01000053.1_96407_96884 100 0.1 0.56 0.03 5 0 YpPGM_8F YpPGM_8-158 500 0.8 4.45
0.04 7.9 0 YpPGM_158R YpPGM_8-158 500 0.8 4.45 0.04 7.9 0 YpPGM_31Fv2 YpPGM_31-205
50 0.05 0.28 0.03 5 0 YpPGM_205Rv2 YpPGM_31-205 50 0.05 0.28 0.03 5 0 Yp-p1202_42780F-
UT1 Yp-p1202_42780-43194 500 0.2 1.11 0.01 2 0 Yp-p1202_43194R-UT2 Yp-p1202_42780-
43194 500 0.2 1.11 0.01 2 0 Yp-p1202_126386F-UT1 Yp-p1202_126386-126750 500 0.2 1.11 0.01
2 0 Yp-p1202_126750R-UT2 Yp-p1202_126386-126750 500 0.2 1.11 0.01 2 0 Yp-
p1202_156402F2-UT1 Yp-p1202_156402-156711 500 0.2 1.11 0.01 2 0 Yp-p1202_156711R-UT2
Yp-p1202_156402-156711 500 0.2 1.11 0.01 2 0 Ft&NN1_F Ftnovicida_CP009607.1 500 0.2 1.11
0.01 2 0 Ft&NN1_R Ftnovicida_CP009607.1 500 0.2 1.11 0.01 2 0 Ft&NN2_F
Ftularensis_CP000915.1_1782 100 0.1 0.56 0.03 5 0 Ft&NN2_R Ftularensis_CP000915.1_1782
100 0.1 0.56 0.03 5 0 Ft&NN3_F Ftularensis_CP000915.1_731 500 1.6 8.9 0.08 15.9 0
Ft&NN3_R Ftularensis_CP000915.1_731 500 1.6 8.9 0.08 15.9 0 Ft&NN4_F
Fnoatunensis_CP003402.1_1749 500 0.2 1.11 0.01 2 0 Ft&NN4_R

*Fnoatunensis*_CP003402.1_1749 500 0.2 1.11 0.012 0 Ft&NN5_F
*Fnoatunensis*_CP003402.1_424 500 0.2 1.11 0.01 2 0 Ft&NN5_R *Fnoatunensis*_CP003402.1_424
 500 0.2 1.11 0.01 2 0 Ft&NN6_F *Fphilom*_CP009444.1_569 500 0.2 1.11 0.01 2 0 Ft&NN6_R
*Fphilom*_CP009444.1_569 500 0.2 1.11 0.01 2 0 Ft&NN7_F *Fphilom*_CP009444.1_285 500 0.2
 1.11 0.01 2 0 Ft&NN7_Rv2 *Fphilom*_CP009444.1_285 500 0.2 1.11 0.01 2 0 Ft&NN8_F
*Ft_dup*_CP000915.1_197 100 0.05 0.28 0.01 2.5 0 Ft&NN8_R *Ft_dup*_CP000915.1_197 100 0.05
 0.28 0.01 2.5 0 Ft&NN9_F *FtLVS*_AM233362_1646546 100 0.05 0.28 0.01 2.5 0 Ft&NN9_R
*FtLVS*_AM233362_1646546 100 0.05 0.28 0.01 2.5 0 Ft&NN10_F *FtLVS*_AM233362_1643765
 100 0.1 0.56 0.03 5 0 Ft&NN10_R *FtLVS*_AM233362_1643765 100 0.1 0.56 0.03 5 0
 Ft&NN11_F *FtLVS*_AM233362_1562618 100 0.1 0.56 0.03 5 0 Ft&NN11_R
*FtLVS*_AM233362_1562618 100 0.1 0.56 0.03 5 0 9F_FtA1_UT1 FtA1 100 0.1 0.56 0.03 5 0
 246R_FtA1_UT2 FtA1 100 0.1 0.56 0.03 5 0 34F_FtA2_UT1 FtA2 100 0.1 0.56 0.03 5 0
 169R_FtA2_UT2 FtA2 100 0.1 0.56 0.03 5 0 28F_FtB_UT1 FtB 100 0.1 0.56 0.03 5 0
 173R_FtB_UT2 FtB 100 0.1 0.56 0.03 5 0 14F_FtA_UT1 FtA 100 0.1 0.56 0.03 5 0
 207R_FtA_UT2 FtA 100 0.1 0.56 0.03 5 0 IPSC-f2 IPSC 20 0.03 0.14 0.03 6.2 0 IPSC-r2 IPSC 20
 0.03 0.14 0.03 6.2 0 580.6 uL total volume of primers 310.4 uL of 1x TE to add to bring up to
 desired volume of primer mix stock 891.0 Total (uL)

(118) TABLE-US-00023 TABLE 11 *Burkholderia* primers and primers with
 Universal Tail (UT). The UT sequence is underlined. Target SEQ SEQ Assay species/
 ID ID name gene NO: NO: Forward Forward sequence primer name (5'->3') Forward
 sequence w/UT TTS1_ TTS1 BpAmpSeq_1 TCGTCGTCACCGGGATGGTC 61
ACCCAACTGAATGGAGCTCGTCG 307 BPSS1_F TCACCGGGATGGTC 407 Reverse
 Reverse sequence primer name (5'->3') Reverse sequence w/UT BpAmpSeq_1
 GGCCTTTGCCCGCATACTCG 62 ACGCACTTGACTTGTCTTCGGCC 308_R
 TTTGCCCGCATACTCG Forward Forward sequence primer name (5'->3') Forward
 sequence w/UT LXCCO *pseudomallei* BpAmpSeq_3 TCGCAWGAAGTGCGTTGCC 63
ACCCAACTGAATGGAGCTCGCA 309 100014_F G WGAAGTGCGTTGCCG 1.1_392
 Reverse Reverse sequence 96_398 primer name (5'->3') Reverse sequence w/UT 17
 BpAmpSeq_3 GCCGCTTGCGAAGCGATGAT 64 ACGCACTTGACTTGTCTTCGCCG 310
 _R CTTGCGAAGCGATGAT Forward Forward sequence primer name (5'->3') Forward
 sequence w/UT LXBY0 *pseudomallei* BpAmpSeq_4 CGCGCTTGCCCAACTACCAG 65
ACCCAACTGAATGGAGCCGCGCT 311 100008_F TGCCCAACTACCAG 7.1_757 Reverse
 Reverse sequence 60_767 primer name (5'->3') Reverse sequence w/UT 51 BpAmpSeq_4
 GCGCAACGGTGCGAGACAA 66 ACGCACTTGACTTGTCTTCGCGC 312_R T
 AACGGTGCGAGACAAT Forward Forward sequence primer name (5'->3') Forward
 sequence w/UT LXCD0 *pseudomallei* BpAmpSeq_5 AATCCATGCATGTCGYGCCC 67
ACCCAACTGAATGGAGCAATCCA 313 100000_F TGCATGTCGYGCCC 2.1_996 Reverse
 Reverse sequence 52_100 primer name (5'->3') Reverse sequence w/UT 245 BpAmpSeq_5
 GCGATCGCTCAACGGGCTTC 68 ACGCACTTGACTTGTCTTCGCGA 314_R
 TCGCTCAACGGGCTTC Forward Forward sequence primer name (5'->3') Forward
 sequence w/UT LXCE0 *pseudomallei* BpAmpSeq_6 TCGCATTTGCAYACGCTCCC 69
ACCCAACTGAATGGAGCTCGCAT 315 100012_F TTGCAYACGCTCCC 3.1_342 Reverse
 Reverse sequence 20_347 primer name (5'->3') Reverse sequence w/UT 47 BpAmpSeq_6
 AGTGCGCAAACCTTGGCGAG 70 ACGCACTTGACTTGTCTTCAGTG 316_R G
 CGCAAACCTTGGCGAGG Forward Forward sequence primer name (5'->3') Forward
 sequence w/UT LWWC *pseudomallei* BpAmpSeq_8 CCTTTGCGGCAAGCGTCGAA 81
ACCCAACTGAATGGAGCCCTTTG 317 010001 *mallei*_F CGGCAAGCGTCGAA 87.1 18
 Reverse Reverse sequence primer name (5'->3') Reverse sequence w/UT BpAmpSeq_8
 GAGCCAACGCACATGGACG 82 ACGCACTTGACTTGTCTTCGAGC 318_R G
 CAACGCACATGGACGG Forward Forward sequence primer name (5'->3') Forward

sequence w/UT LBW *pseudomallei* BpAmpSeq_1 CCAGTCGGGCGGAAAAA 83
ACCCAACTGAATGGAGCCCAGTC 319 010001 *mallei* 0_F C GGGCCGGGAAAAAAC 25.1_17
Reverse Reverse sequence 183_17 primer name (5'→3') Reverse sequence w/UT 602
BpAmpSeq_1 GGCGGCAAAAGCGTCGATG 84 ACGCACTTGACTTGTCTTCGGCG 320
0_R A GCAAAAGCGTCGATGA Forward Forward sequence primer name (5'→3') Forward
sequence w/UT LXAY0 *pseudomallei* BpAmpSeq_1 GCCGGAACCGTCGAGCATT 85
ACCCAACTGAATGGAGCGCCGG 321 100036 *mallei* 1_F G AACCGTCGAGCATTG 7.1_0_6
Reverse Reverse sequence 40 primer name (5'→3') Reverse sequence w/UT BpAmpSeq_1
TGGATTCGACTGCCTCCGCT 86 ACGCACTTGACTTGTCTTCTGGA 322 1_R
TTCGACTGCCTCCGCT Forward Forward sequence primer name (5'→3') Forward
sequence w/UT LWVY *pseudomallei* BpAmpSeq_1 TCGATATCCGCCGTCTCGCC 87
ACCCAACTGAATGGAGCTCGATA 323 010001 *mallei* 2_F TCCGCCGTCTCGCC 90.1_17
Reverse Reverse sequence 226_17 primer name (5'→3') Reverse sequence w/UT 689
BpAmpSeq_1 ATGTGTCGGTGGGCTTCGGT 88 ACGCACTTGACTTGTCTTCATGT 324
2_R GTCGGTGGGCTTCGGT Forward Forward sequence primer name (5'→3') Forward
sequence w/UT LXAD0 *pseudomallei* BpAmpSeq_1 GAAAGGCGATGTGCCGAGC 89
ACCCAACTGAATGGAGCGAAAG 325 100005 *mallei* 3_F G GCGATGTGCCGAGCG 9.1_247
Reverse Reverse sequence 60_250 primer name (5'→3') Reverse sequence w/UT 75
BpAmpSeq_1 TTCGGAGAAGCGCCAAACG 90 ACGCACTTGACTTGTCTTCTTCGG 326
3_R C AGAAGCGCCAAACGC Forward Forward sequence primer name (5'→3') Forward
sequence w/UT NC_00 *pseudomallei* BpAmpSeq_1 GCCAGCGCATCCACCAACAT 109
ACCCAACTGAATGGAGCGCCAGC 327 6350_2 complex SNP 6_F GCATCCACCAACAT
289827 Reverse Reverse sequence primer name (5'→3') Reverse sequence w/UT
BpAmpSeq_1 AGAGGAAGAAGGGCGAGGC 110 ACGCACTTGACTTGTCTTCAGAG 328
6_R G GAAGAAGGGCGAGGCG Forward Forward sequence primer name (5'→3')
Forward sequence w/UT NC_00 *cepacia* BpAmpSeq_1 CGCGCARYTCGTCGTCCTCG 111
ACCCAACTGAATGGAGCCGCGCA 329 6350_1 complex 8_F RYTCGTCGTCCTCG 33027
SNPs Reverse Reverse sequence primer name (5'→3') Reverse sequence w/UT
BpAmpSeq_1 CGAACCTSGTGCMGGTRCA 112 ACGCACTTGACTTGTCTTCCGAA 330 8_R
G CCTSGTGCMGGTRCAG Forward Forward sequence primer name (5'→3') Forward
sequence w/UT NC_00 Bpc MSS BpAmpSeq_1 CACGTTGCCSGGRAARTACG 113
ACCCAACTGAATGGAGCCACGTT 331 6350_2 9_F GCCSGGRAARTACG 248145- Reverse
Reverse sequence 224819 primer name (5'→3') Reverse sequence w/UT 3 BpAmpSeq_1
CCGTCGACAAGATCGCGCTS 114 ACGCACTTGACTTGTCTTCCCGTC 332 9_R
GACAAGATCGCGCTS Forward Forward sequence primer name (5'→3') Forward
sequence w/UT NC_00 Bpc MSS BpAmpSeq_2 CAGAACGCGCTRTYCCACG 115
ACCCAACTGAATGGAGCCAGAA 333 6350_9 0_F CGCGCTRTYCCACG 88041- Reverse
Reverse sequence 988089 primer name (5'→3') Reverse sequence w/UT BpAmpSeq_2
TGCCGCGTGATCCATTGCAT 116 ACGCACTTGACTTGTCTTCTGCC 334 0_R
GCGTGATCCATTGCAT Forward Forward sequence primer name (5'→3') Forward
sequence w/UT Bm_11 *mallei* BpAmpSeq_2 AGGGGGTGGTTTCTGAGTG 117
ACCCAACTGAATGGAGCAGGGG 335 589 1_F GCGTGAC
GTGGTTTCTGAGTGGCGTGAC Reverse Reverse sequence primer name (5'→3')
Reverse sequence w/UT BpAmpSeq_2 AGCGGTGTCGACGGGTGGA 118
ACGCACTTGACTTGTCTTCAGCG 336 1_R AAGGATG GTGTCGACGGGTGGAAGGATG
Forward Forward sequence primer name (5'→3') Forward sequence w/UT Bm_11 *mallei*
BpAmpSeq_2 ACGGGCGCTTCACGATCTCG 119 ACCCAACTGAATGGAGCACGGG 337
767 2_F GTGTTC CGCTTCACGATCTCGGTGTTC Reverse Reverse sequence primer name
(5'→3') Reverse sequence w/UT BpAmpSeq_2 GCGCGGCAGTTCGATCAGG 120
ACGCACTTGACTTGTCTTCGCGC 338 2_R CATTG GGCAGTTCGATCAGGCATTG

Forward Forward sequence primer name (5'->3') Forward sequence w/UT PBP3- pbp3
BpAmpSeq_2 ATCCGCCGTCCCGCCCAGCA 305 ACCCAACTGAATGGAGCATCCGC 339
170- 4_F ATAG CGTCCCGCCCAGCAATAG 505 Reverse Reverse sequence primer name (5'->3') Reverse sequence w/UT BpAmpSeq_2 GGGTTCGCCAGATTTCGTA 306
ACGCACTTGACTTGTCTTCGGGT 340 4_R GGTGGTGAG
TCGCCAGATTTCGTAGGTGGTG AG Forward Forward sequence primer name (5'->3')
Forward sequence w/UT K9pen penA BpAmpSeq_2 CGGTCGCCACAAATTCGCAC 127
ACCCAACTGAATGGAGCCGGTCG 341 A378- 6_F GCACTC
CCACAAATTCGCACGCACTC 529 Reverse Reverse sequence primer name (5'->3')
Reverse sequence w/UT BpAmpSeq_2 AGCGAGCGGCGCAACGGAG 128
ACGCACTTGACTTGTCTTCAGCG 342 6_R AATGATT AGCGGCGCAACGGAGAATGATT
Forward Forward sequence primer name (5'->3') Forward sequence w/UT K9pen penA
BpAmpSeq_2 GCTGCGCGGCCAAGCGAAA 129 ACGCACTTGACTTGTCTTCGCTG 343
A575- 7_F AACG CGCGGCCAAGCGAAAAACG 761 Reverse Reverse sequence primer
name (5'->3') Reverse sequence w/UT BpAmpSeq_2 CGCGAGGACCGCAGCGCAA 130
ACCCAACTGAATGGAGCCGCGA 344 7_R AGC GGACCGCAGCGCAAAGC Forward
Forward sequence primer name (5'->3') Forward sequence w/UT K9pen penA
BpAmpSeq_2 GGCCGCAGACCGTCACCGC 131 ACCCAACTGAATGGAGCGGCCGC 345
A949- 8_F GTATG AGACCGTCACCGCGTATG 1172 Reverse Reverse sequence primer
name (5'->3') Reverse sequence w/UT BpAmpSeq_2 GTCGCCCGTCTTGTGTCGA 132
ACGCACTTGACTTGTCTTCGTCG 346 8_R GCATC CCCGTCTTGTGTCGAGCATC
(119) TABLE-US-00024 TABLE 12 *Bacillus* primers and primers with Universal
Tail (UT). The UT sequence is underlined. SEQ SEQ ID ID NO: NO: Target Forward
Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3') w/UT
plcR pIcR Ba- TTTTTCGTAAGCATCTTCAA 29 ACCCAACTGAATGGAGCTTTTTC 347
specific- GTAAGCATCTTCAA 1F_UT1 Reverse Reverse Reverse primer sequence sequence
name (5' -> 3') w/UT Ba- TTTGATGTGAAGGTGAGACA 30
ACGCACTTGACTTGTCTTCTTTGA 348 specific- TGTGAAGGTGAGACA IRUT2 Target
Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
w/UT CP0088 Ba- ACGTCAGGTGATTATTGGAC 1 ACCCAACTGAATGGAGCACGTCA 349
53.1_ specific- GGTGATTATTGGAC 5309 3F_UT1 Reverse Reverse Reverse primer sequence
sequence name (5' -> 3') w/UT Ba- CAACAATTATATCCGCCATT 2
ACGCACTTGACTTGTCTTCCAAC 350 specific- AATTATATCCGCCATT 3RUT2 Target
Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
w/UT CP(K)88 Ba- GAAGATGTACGCTCGATAG 3 ACCCAACTGAATGGAGCGAAGA 351
53.1_ specific- G TGTACGCTCGATAGG 5316 5FUT1 Reverse Reverse Reverse primer sequence
sequence name (5' -> 3') w/UT Ba- GAAATTCTTTTGGCCATCAC 4
ACGCACTTGACTTGTCTTCGAAA 352 specific- TTCTTTTGGCCATCAC 5RUT2 Target
Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
w/UT CP0127 Ba- CACAATTGAATGAAAATGCT 5 ACCCAACTGAATGGAGCCACAAT 353
25.1_ specific- TGAATGAAAATGCT 3629 6F_UT1 Reverse Reverse Reverse primer sequence
sequence name (5' -> 3') w/UT Ba- CACGAAACCTGTTTACCTTT 6
ACGCACTTGACTTGTCTTCCACG 354 specific- AAACCTGTTTACCTTT 6RUT2 Target
Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
w/UT CP0127 Ba- GATATTCGACGAGCTTTCTG 7 ACCCAACTGAATGGAGCGATATT 355
25.1_ specific- CGACGAGCTTTCTG 5103 8F_UT1 Reverse Reverse Reverse primer sequence
sequence name (5' -> 3') w/UT Ba- TATTCATCGTCATCCTCCTC 8
ACGCACTTGACTTGTCTTCTATT 356 specific- CATCGTCATCCTCCTC 8R_UT2 Target
Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
w/UT CP0127 Ba- TATTGAACGCATTGAATCAG 9 ACCCAACTGAATGGAGCTATTGA 357

25.1_ specific- ACGCAATGATCAG 5107 9F_UT1 Reverse Reverse primer sequence
 sequence name (5' -> 3') w/UT Ba- TATTGGTAAGCAAACCGTCT 10
ACGCACTTGACTTGTCTTCTATTG 358 specific- GTAAGCAAACCGTCT 9RUT2 Target
 Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
 w/UT JSZQ01 Ba- GGTTCAGGACAAAATGTAG 11 ACCCAACTGAATGGAGCGGTTCA 359
 000034. specific- C GGACAAAATGTAGC 1_220 11F_UT1 Reverse Reverse Reverse primer
 sequence sequence name (5'-> 3') w/UT Ba- TAACTTCTGAAGCGAAAACC 12
ACGCACTTGACTTGTCTTCTTA ACT 360 specific- TCTGAAGCGAAAACC 11R_UT2 Target
 Forward Forward Forward Assay species/ primer sequence sequence name gene name (5' -> 3')
 w/UT JSZS010 Ba- GCGAATTTTAGACGACAATC 13 ACCCAACTGAATGGAGCGCGAAT
 361 00036.1_5 specific- TTTAGACGACAATC 12F_UT1 Reverse Reverse Reverse primer
 sequence Sequence name (5' -> 3') w/UT Ba- TAACCGTGCTTAATTCGTTT 14
ACGCACTTGACTTGTCTTCT 362 specific- AACCGTGCTTAATTCGTTT 12RUT2 Target
 Forward Forward Forward Assay species/ primer sequence Sequence name gene name (5' -> 3')
 w/UT LGCC0 Ba- ATTAATAAGGCGACTGGTGA 15 ACCCAACTGAATGGAGCATT 363
 1000010.1_ specific- AATAAGGCGACTGGTGA 232 14F_UT1 Reverse Reverse Reverse primer
 sequence Sequence name (5' -> 3') w/UT Ba- TTACCCATCCAGAATGAGAC 16
ACGCACTTGACTTGTCTTCT 364 specific- TACCCATCCAGAATGAGAC 14RUT2 Target
 Forward Forward Forward Assay species/ primer sequence Sequence name gene name (5' -> 3')
 w/UT LGCC0 Ba- ACAATTCTTAAAAGGCGACA 17 ACCCAACTGAATGGAGCACA 365
 1000048.1_ specific- ATTCTrAAAAGGCGACA 280 16F_UT1 Reverse Reverse Reverse primer
 sequence Sequence name (5' -> 3') w/UT Ba- TGTAGCGTCTCCGATATTTT 18
ACGCACTTGACTTGTCTTCT 366 specific- GTAGCGTCTCCGATATTTT 16R_UT2 Assay
 Target Forward Forward sequence Forward sequence name species/gene primer name (5' -
 > 3') w/ UT NNLO Ba- CATGGGGCTTTCTATTATGT 19
ACCCAACTGAATGGAGCCATGGG 367 MU010000 specific- GCTTTCTATTATGT 90.1_49
 20FUT1 Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/ UT Ba-
 TTCGTTCTTTTCATAAGTTTCC 20 ACGCACTTGACTTGTCTTCTTTCGT 368 specific- T
 TCTTTTCATAAGIF1CCT 20R_UT2 Assay Target Forward Forward sequence Forward
 sequence name species/gene primer name (5' -> 3') w/UT NN_LO Ba-
 TTGGAGTITGTTITGCTTTT 21 ACCCAACTGAATGGAGCTTGAG 369 QC 0100 specific-
 TTTGTTTTTGCTTTT 0013.1_3 22F_UT1 Reverse Reverse sequence Reverse sequence
 w/UT primer name (5' -> 3') Ba-specific- GTAACAATTAATCCACGTCC 22
ACGCACTTGACTTGTCTTCTCGTAA 370 22Rv2_UT2 T CAATTAATCCACGTCCT Assay Target
 Forward Forward sequence Forward sequence name species/gene primer name (5' -> 3')
 w/UT pX01 pX01 pX01_ TGAGCCTACCTAGTGATTGG 33
ACCCAACTGAATGGAGCTGAGCC 371 113F_UT1 TACCTAGTGATTGG Reverse Reverse
 sequence Reverse sequence primer name (5' -> 3') w/ UT pX01-
 TTGGATAAATTCCACAAATT 34 ACGCACTTGACTTGTCTTCTTGG 372 315Rv2_UT2
 CCTC ATAAATTCCACAAATTCCTC Assay Target Forward Forward sequence Forward
 sequence name species/gene primer name (5' -> 3') w/UT pX02 pX02 pX02_101F_
 CGCCAGCGTATTATATAGGT 35 ACCCAACTGAATGGAGCCGCCAG 373 UT1
 CGTATTATATAGGT Reverse primer Reverse sequence Reverse sequence w/UT name (5' -
 > 3') pX02 269R GCTAATTCTGGGTTGTGTTT 36 ACGCACTTGACTTGTCTTCTCGCTA 374
 UT2 ATTCTGGGTTGTGTTT Assay Target Forward Forward sequence name species/gene
 primer name (5' -> 3') Forward sequence w/UT gyrA gyrA gyrA_28Fv2_
 TCGGTAAGTATCACCTCA 37 ACCCAACTGAATGGAGCTCGGTA 375 UT1
 AGTATCACCTCA Reverse primer Reverse sequence name (5' -> 3') Reverse sequence
 w/ UT gyrA_182Rv2 TGCTTCTGTATAACGCATT 38 ACGCACTTGACTTGTCTTCTGCTT
 376 _UT2 CTGTATAACGCATT Assay Target Forward Forward sequence Forward sequence

name species/gene primer name (5' -> 3') w/UT parC parC_1F_UTI
 CAGTCGGTAACGTTATTGGT 39 ACCCAACTGAATGGAGCCAGTCG 377
 GTAACGTTATTGGT Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/UT parC_197R_U TAACTCAGATGCAATTGGTG 40
ACGCACTTGACTTGTCTTCTTAAC 378 T2 CAGATGCAATTGGTG Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT
 gyrB gyrB gyrB_8F_UT ATTGTAGAGGGGTGACTCTGC 41
ACCCAACTGAATGGAGCATTGTA 379 1 GAGGGTGACTCTGC Reverse Reverse sequence
 Reverse sequence primer name (5' -> 3') w/UT gyrB_194R_ TATCAAAATCTCCGCCAAT
 42 ACGCACTTGACTTGTCTTCTTATCA 380 UT2 AAATCTCCGCCAAT Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT
 pagA pagA 801F_pagAv3_ GGTTACAGGACGGATTGAT 31
ACCCAACTGAATGGAGCGGTTAC 381 UT1 A AGGACGGATTGATA Reverse Reverse
 sequence Reverse sequence primer name (5' -> 3') w/UT 1042R_pagAv
 TCCCACCAATATCAAAGAAC 32 ACGCACTTGACTTGTCTTCTCCCA 382 3UT2
 CCAATATCAAAGAAC Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT rpoB rpoB rpoB_29F_U
 TTCTTCGGAAGTTCTCAGTT 43 ACCCAACTGAATGGAGCTTCTTC 383 T1
 GGAAGTTCTCAGTT Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/UT rpoB_196R_ CGGACACATACGACCATAG 44 ACGCACTTGACTTGTCTTCCGGA
 384 UT2 CACATACGACCATAG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT AA_2502 AA_2502_UT
 AAGTTTGAGGTGTGGAAAT 45 ACCCAACTGAATGGAGCAAGTTT 385 1 G
 GAGGTGTGGAAATG Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/UT AA_2502_UT TCGAAATGAGTTCCAATTTT 46 ACGCACTTGACTTGTCTTCTCGA
 386 2 AATGAGTTCCAATTTT Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT AA_2503 AA_2503v2_
 CAAAATAATAGGGGAGGG 47 ACCCAACTGAATGGAGCCAAAA 387 UT1 TG
 CTAATAGGGGAGGGTG Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/UT AA_2503_UT CCGAGAACCTACCTCGTTA 48
ACGCACTTGACTTGTCTTCCCGA 388 2 GAACCTACCTCGTTA Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT
 Ba_AmesAnc_ Ba&NN32_F AGGAGATGAGAGTTTTGCA 49
ACCCAACTGAATGGAGCAGGAG 389 4669915 C ATGAGAGTTTTGCAC Reverse Reverse
 sequence Reverse sequence primer name (5' -> 3') w/UT Ba&NN32_R
 ACCCCCATAATTACCATGA 50 ACGCACTTGACTTGTCTTCACCC 390
 CCATAATTACCATGA Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT Ba_AmesAnc_ Ba&NN33_F
 CGTTGCGTAAGTATGTGCTA 51 ACCCAACTGAATGGAGCCGTTGC 391 4001578
 GTAAGTATGTGCTA Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/UT Ba&NN33_R AGGTGGCGTAATTAACGTA 52 ACGCACTTGACTTGTCTTCAGGT
 392 G GGCGTAATTAACGTAG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT Ba_AmesAnc_ Ba&NN37_F
 CGAAAAGTTGTCGACCTAAT 53 ACCCAACTGAATGGAGCCGAAA 393 1069024
 AGTTGTCGACCTAAT Reverse Reverse sequence Reverse sequence primer name (5' -> 3') w/UT Ba&NN37_R ACTGCGTTCACGAAGAATA 54 ACGCACTTGACTTGTCTTCACTG
 394 G CGTTCACGAAGAATAG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT Ba_AmesAnc_ Ba&NN38_F
 TCTCTTGATTCAACGTTTCC 55 ACCCAACTGAATGGAGCTCTCTT 395 3668548
 GATTCAACGTTTCC Reverse Reverse sequence Reverse sequence primer name (5' ->

3') w/UT Ba&NN38_R GATGCAAAACCAATTCATT 56 ACGCACTTGACTTGTCTTCGATG
396 CAAAACCAATTCACCTT Assay Target Forward Forward sequence Forward sequence
name species/gene primer name (5' -> 3') w/UT Ba_AmesAnc_Ba&NN40_F
GTGAAACATCGCTTTTATAGG 57 ACCCAAGTGAATGGAGCGTGAA 397 371913
ACATCGCTTTTATAGG Reverse Reverse sequence Reverse sequence primer name (5' ->
3') w/UT Ba&NN40_R TCCGCAATGATATACTTCAA 58
ACGCACTTGACTTGTCTTCTCCGC 398 AATGATATACTTCAA Assay Target Forward
Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT
BaA_mesAnc_Ba&NN41_F ATACGGTGAAAATGAAGCA 59
ACCCAAGTGAATGGAGCATAACGG 399 999035 G TGAAAATGAAGCAG Reverse Reverse
sequence Reverse sequence primer name (5' -> 3') w/UT Ba&NN41 R
CGTCTTTGGTAATCGTTCA 60 ACGCACTTGACTTGTCTTCCGTCT 400
TTGGTAATCGTTCA Assay Target Forward Forward sequence Forward sequence name
species/gene primer name (5' -> 3') w/UT ChimpKiller_ChimpKiller_
TTATCGTCCATTCTTTCGTC 23 ACCCAAGTGAATGGAGCTTATCG 401 9-159 9F
TCCATTCTTTCGTC Reverse Reverse sequence Reverse sequence primer name (5' ->
3') w/UT ChimpKiller_AAACCTAATGAAACGGGAT 24 ACGCACTTGACTTGTCTTCAAAC
402 159R T CTAATGAAACGGGATT Assay Target Forward Forward sequence Forward
sequence name species/gene primer name (5' -> 3') w/UT ChimpKiller_ChimpKiller_
TATGAAAGGAGCCGTAAAA 25 ACCCAAGTGAATGGAGCTATGAA 403 91-320 91F C
AGGAGCCGTAAAAAC Reverse Reverse sequence Reverse sequence primer name (5' ->
3') w/UT ChimpKiller_TGAATATGAAGCGGAAAAC 26 ACGCACTTGACTTGTCTTCTGAA
404 320R T TATGAAGCGGAAAAC Assay Target Forward Forward sequence Forward
sequence name species/gene primer name (5' -> 3') w/UT ChimpKiller_ChimpKiller_
TCGAACATACCTCCATTCT 27 ACCCAAGTGAATGGAGCTCGAAC 405 481-698 481F
ATACCTCCATTCT Reverse Reverse sequence Reverse sequence primer name (5' -> 3')
w/UT ChimpKiller_AAAGATAGCTTTGCACTTGG 28 ACGCACTTGACTTGTCTTCAAAG
406 698R ATAGCTTTGCACTTGG

(120) TABLE-US-00025 TABLE 13 *Yersinia* primers and primers with Universal
Tail (UT). The UT sequence is underlined. SEQ SEQ ID ID NO: NO: Assay Target
Forward Forward sequence Forward sequence name species/gene primer name (5' -> 3')
w/UT Ypestis_LPQY Yp&NN1_F AACAAAGCTAAAACCGAACA 231
ACCCAAGTGAATGGAGCAACAA 407 01000176.1_7 A GCTAAAACCGAACAA Reverse
Reverse sequence Reverse sequence primer name (5' -> 3') w/UT Yp&NN1_R
ATAGCCTCAACTGCTTTTTG 232 ACGCACTTGACTTGTCTTTCATAG 408
CCTCAACTGCTTTTTG Assay Target Forward Forward sequence Forward sequence name
species/gene primer name (5' -> 3') w/UT Wagner_Yp_Yp&NN2_F
GAAAGGAGTGCGGGTAATA 237 ACCCAAGTGAATGGAGCGAAAG 409 pla_F Forward
GGTT GAGTGCGGGTAATAGGTT Reverse Reverse sequence Reverse sequence primer
name (5' -> 3') w/UT Yp&NN2_R GGCCTGCAAGTCCAATATA 238
ACGCACTTGACTTGTCTTCCGCC 410 TGG TGCAAGTCCAATATATGG Assay Target
Forward Forward sequence Forward sequence name species/gene primer name (5' -> 3')
w/UT YenteroFR72 Yp&NN3_F GATGCTTCTGCTATCAGSTT 263
ACCCAAGTGAATGGAGCGATGCT 411 9477.2_623 TCTGCTATCAGSTT Reverse Reverse
sequence Reverse sequence primer name (5' -> 3') w/UT Yp&NN3_R
GTGTGRCTTTGAASTCTTGT 264 ACGCACTTGACTTGTCTTTCGTGT 412
GRCTTTGAASTCTTGT Assay Target Forward Forward sequence Forward sequence name
species/gene primer name (5' -> 3') w/UT YPCO92_NC_00 Yp&NN4_F
ACTCGGGATACTCCATACT 249 ACCCAAGTGAATGGAGCACTCGG 413 3143_113190 G
GATACTCCATACTG Reverse Reverse sequence Reverse sequence primer name (5' ->

3') w/UT Yp&NN4_R CGAAAGCAGTGGTCAATC 250 ACGCACTTGACTTGTCTTCCGAA
 414 AGCAGTGGTCAATC Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT YPC092_NC_00 Yp&NN5_F
 CATGCGCTTTACGTTATATG 251 ACCCAACTGAATGGAGCCATGCG 415 3143_161621
 CTTTACGTTATATG Reverse Reverse sequence Reverse sequence primer name (5' -> 3')
 w/UT Yp&NN5_R GCGTTCTGCACTCTGTCT 252 ACGCACTTGACTTGTCTTCCGCGTT 416
 CTGCACTCTGTCT Assay Target Forward Forward sequence Forward sequence (5' -> 3')
 w/UT name species/gene primer name YPC092_NC_00 Yp&NN6_F
 AGCGACTTCCGTGATAAAG 253 ACCC'AACTGAATGGAGCAGCGA 417 3143_152213
 CTTCCGTGATAAAG Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Yp&NN6_R ACTCAGGATACCGTGTGGT 254 ACGCACTTGACTTGTCTTCACTC
 418 AGGATACCGTGTGGT Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT YPC092_NC_00 Yp&NN7_F
 TTCACGATAATCCCCTAAT 255 ACCCAACTGAATGGAGCTTCACG 419 3143_129539 G
 ATAATCCCCTAATG Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Yp&NN7_R TTCTGTGCTCTGGCTGATA 256 ACGCACTTGACTTGTCTTCTTCTG
 420 TGCTCTGGCTGATA Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT YPC092_NC_00 Yp&NN8_F
 ATTATCTGTGCCCCTTCTTT 257 ACCCAACTGAATGGAGCATTATC 421 3143_91203
 TGTGCCCCTTCTTT Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Yp&NN8_R GGAGTGGATGCCACTAAAC 258 ACGCACTTGACTTGTCTTCCGGAG
 422 TGGATGCCACTAAAC Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT YpC092_NC_00 Yp&NN9_F
 CCTCACACAACAATTCCTACT 259 ACCCAACTGAATGGAGCCCTCAC 423 3143_121812 G
 ACAACAATTCCTACTG Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Yp&NN9_R TTTTTCGACAAATTTAAG 260 ACGCACTTGACTTGTCTTCTTTTTT
 424 G CCGACAAATTTAAGG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT Yp_AL590842.1_ Yp&NN10_F
 AGCATGAAGGTTGCTAAAA 261 ACCCAACTGAATGGAGCAGCATG 425 RX_SNP G
 AAGGTTGCTAAAAAG Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Yp&NN10_R GGTGACTTCAAAACCGTTA 262 ACGCACTTGACTTGTCTTCCGGTG
 426 G ACTTCAAAACCGTTAG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT AGJT01000065. Yp&NN11_F
 CAGTACCGACAAAACCTTC 233 ACCCAACTGAATGGAGCCAGTAC 427 1_0_338
 CGACAAAACCTTC Reverse Reverse sequence Reverse sequence primer name (5' -> 3')
 w/UT Yp&NN11_R TTTACTACTCTGAAAACGA 234 ACGCACTTGACTTGTCTTCTTTAC
 428 G TACTCTGAAAACGAG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT FAURO1000053.1_ Yp&NN12_F
 GCACTACAAATTTAAATCC 235 ACCCAACTGAATGGAGCGCACTA 429 9640_796884 C
 CAAATTTAAATCCC Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Yp&NN12_R GTCGATTATCAACCTCTAT 236 ACGCACTTGACTTGTCTTCCGTCG
 430 G ATTATCAACCTCTATG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT YpPGM_PGM YpPGM_8F
 TTAATATCCCGGCACTCAT 239 ACCCAACTGAATGGAGCTTAATA 431 8-158 A
 TCCCGGCACTCATA Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT YpPGM_158 TCCTTAACTGAATAAGTGC 240 ACGCACTTGACTTGTCTTCTCCTT
 432 R TCAAACTGAATAAGTGCTCA Assay Target Forward Forward sequence Forward
 sequence name species/gene primer name (5' -> 3') w/UT YpPGM_PGM YpPGM_31F
 TTTAATGAACGGTGCCTAG 241 ACCCAACTGAATGGAGCTTTAAT 433 31-205 v2
 GAACGGTGCCTAG Reverse Reverse sequence Reverse sequence primer name (5' ->

3') w/UT YpPGM_205 GTCTGCGTTTCTCCAGTAT 242 ACGCACTTGACTTGCTCTCGTCTG
 434 Rv2 CGTTTCTCCAGTAT Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT Yp-p1202_ plP1202 Yp-
 TCTGGCCTGCTAAATAAAA 243 ACCCAACTGAAIGGAGCTCTGGC 435 42780-
 P1202_42780 ACGAACC CTGCTAAATAAAAACGAACC 43194 F-UT1 Reverse Reverse
 sequence Reverse sequence primer name (5' -> 3') w/UT Yp-
 CAGGCCTCAGCATTTTATT 244 ACGCACTTGACTTGCTCTTCCAGG 436 p1202_
 ATGGTGAT CCTCAGCATTTTATTATGGTGAT 43194 R-UT2 Assay Target Forward Forward
 sequence Forward sequence name species/gene primer name (5' -> 3') w/UT Yp- plP1202
 Yp- GGGGCGGATACCTTCACCT 245 ACCCAACTGAATGGAGCGGGGC 437 p1202_1
 P1202_ ATG GGATACCTTCACCTATG 26386- 12638 126750 6F-UT1 Reverse Reverse
 sequence Reverse sequence primer name (5' -> 3') w/UT Yp-
 CTGGGGTTCAGTCTGGACG 246 ACGCACTTGACTTGCTCTTCCTGG 438 p1202_ AGAT
 GGTTCACTCTGGACGAGAT 126750R- UT2 Assay Target Forward Forward sequence
 Forward sequence name species/gene primer name (5' -> 3') w/UT Yp-p1202_ plP1202 Yp-
 ACCATCCGCGCTAAATCG 247 ACCCAACTGAATGGAGCACCATC 439 156402- p1202_
 TC CGGCGCTAAATCGTC 156711 15640 2F2-UT1 Reverse Reverse sequence Reverse
 sequence primer name (5' -> 3') w/UT Yp- GAAATGCGCCTGGTAAGCA 248
ACGCACTTGACTTGCTCTTCGAAA 440 pl202_15671 GAGT TGCGCCTGGTAAGCAGAGT
 1R-UT2
 (121) TABLE-US-00026 TABLE 14 Francisella primers and primers with Universal
 Tail (UT). The UT sequence is underlined. SEQ SEQ ID ID NO: NO: Assay Target
 Forward Forward sequence Forward sequence name species/gene primer name (5' -> 3')
 w/UT Ftnovida_ Ft&NN1_F GGTAGGATAACTACCAAG 285
ACCCAACTGAATGGAGCGGTAG 441 CP009607.1 GATAACTACCAAG Reverse Reverse
 sequence Reverse sequence primer name (5' -> 3') w/UT Ft&NN1_R
 GTCATGAGTTTTACCAATA 286 ACGCACTTGACTTGCTCTTCGTCAT 442 CTC
 GAGTTTTACCAATACTC Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT Ftularensis_CP0 Ft&NN2_F
 GAAGTGGCTCATGTTAGAG 265 ACCCAACTGAATGGAGCGAAGT 443 00915.1_1782 G
 GGCTCATGTTAGAGG Reverse Reverse sequence Reverse sequence primer name (5' -
 > 3') w/UT Ft&NN2_R AGCGAGCCTATATGTAACC 266
ACGCACTTGACTTGCTTCAGCG 444 A AGCCTATATGTAACCA Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT
 Ftularensis_ Ft&NN3_F TTTAATGTCCGTCAACCTCT 267
ACCCAACTGAATGGAGCTTTAAT 445 CP000915.1_731 GTCCGTCAACCTCT Reverse
 Reverse sequence Reverse sequence primer name (5' -> 3') w/UT Ft&NN3_R
 ACGAGTTTGTGAGTCGCTA 268 ACGCACTTGACTTGCTTCACGA 446 T
 GTTTGTGAGTCGCTAT Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT Fnoatunensis_CP Ft&NN4_F
 CGGTAAGAATACGACCAGA 291 ACCCAACTGAATGGAGCCGGTAA 447 003402.1_1749 G
 GAATACGACCAGAG Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Ft&NN4_R AGAGGATTTCTTCCTCCTTG 292 ACGCACTTGACTTGCTTCAGAG
 448 GATTTCTTCCTCCTTG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT Fnoatunensis_ Ft&NN5_F
 AATTCTACAAGCACCTGGA 293 ACCCAACTGAATGGAGCAATTCT 449 CP003402.1_424
 A ACAAGCACCTGGAA Reverse Reverse sequence Reverse sequence primer name (5' -
 > 3') w/UT Ft&NN5R TCCTATTAAAAGCGCCATA 294
ACGCACTTGACTTGCTTCCTCCTA 450 G TTAAGCGCCATAG Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT

Fphilom_CP009 Ft&NN6F CTTATGCAGAGAGGAAC 287
ACCCAACTGAATGGAGCCTTATG 451 444.1_569 T CAGCAAGAGGAACT Reverse
 Reverse sequence Reverse sequence primer name (5' -> 3') w/UT Ft&NN6_R
 ATACACCGGGATAGGTTTC 288 ACGCACTTGACTTGTCTTCATAC 452 T
 ACCGGGATAGGTTTCT Assay Target Forward Forward sequence Forward sequence name
 species/gene primer name (5' -> 3') w/UT Fphilom_CP009 Ft&NN7_F
 CTGATGGAAGAGAGTTCGA 289 ACCCAACTGAATGGAGCCTGATG 453 444.1_285 G
 GAAGAGAGTTTCGAG Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Ft&NN7_Rv2 GTAGATATAATCAGCGCCA 290 ACGCACTTGACTTGTCTTCCGTAG
 454 C ATATAATCAGCGCCAC Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT Ft_dup_CP0009 Ft&NN8_F
 TGTTACGTACAGGCTGTCA 269 ACCCAACTGAATGGAGCTGTTAC 455 15.1_197 A
 GTACAGGCTGTCAA Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Ft&NN8_R ATCATATCCCGTAGCACAA 270 ACGCACTTGACTTGTCTTCATCAT
 456 G ATCCCGTAGCACAAAG Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT FtLVS_AM2333 Ft&NN9_F
 ATCAAGCTCATCTTCAAAG 279 ACCCAACTGAATGGAGCATCAAG 457 62_1646546 C
 CTCATCTTCAAAGC Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Ft&NN9_R AACCATGTTTCAGATCCAAA 280 ACGCACTTGACTTGTCTTCCAACC
 458 A ATGTTTCAGATCCAAAA Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT FtLVS_AM2333 Ft&NN10_F
 TACCTCTGCCAAAAATTCA 281 ACCCAACTGAATGGAGCTACCTC 459 62_1643765 T
 TGCCAAAAATTCAT Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Ft&NN10_R GGCATACTCAAGGTAGTGG 282 ACGCACTTGACTTGTCTTCGGCA
 460 T TACTCAAGGTAGTGGT Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT FtLVS_AM2333 Ft&NN11_F
 TCTTTGGTAGCTTGCTGACT 283 ACCCAACTGAATGGAGCTCTTTG 461 62_1562618
 GTAGCTTGCTGACT Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT Ft&NN11_R CAGACGACACTTGGCTTAT 284 ACGCACTTGACTTGTCTTCCAGA
 462 T CGACACTTGGCTTATT Assay Target Forward Forward sequence Forward sequence
 name species/gene primer name (5' -> 3') w/UT FtA1 FtA1 9F_FtA1_UT
 CATAACCCATCGCAATATC 271 ACCCAACTGAATGGAGCCATAAC 463 1 T
 CCATCGCAATATCT Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT 246R_FtA1_AAATTATCTGTAGCGGCAA 272 ACGCACTTGACTTGTCTTCCAAAT
 464 UT2 A TATCTGTAGCGGCAAAA Assay Target Forward Forward sequence Forward
 sequence name species/gene primer name (5' -> 3') w/UT FtA2 FtA2 34F_FtA2_U
 GTGTCCAACGAAACCATAA 273 ACCCAACTGAATGGAGCGTGTCC 465 T1 T
 AACGAAACCATAAT Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT 169R_FtA2_TTTGGTTGATTCTGTCTCAGTG 274
ACGCACTTGACTTGTCTTCTTTGG 466 UT2 TTGATTCTGTCTCAGTG Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT FtB
 FtB 28F_FtB_UT AAGCTTAAGTGGTATTGG 275 ACCCAACTGAATGGAGCAAGCTT 467
 1 A AACTGGTGATTGGA Reverse Reverse sequence Reverse sequence primer name (5' ->
 3') w/UT 173R_FtB_U CGCCTAACATCTTATCTGCT 276
ACGCACTTGACTTGTCTTCCGCCT 468 T2 AACATCTTATCTGCT Assay Target Forward
 Forward sequence Forward sequence name species/gene primer name (5' -> 3') w/UT FtA
 FtA 14F_FtA_UT GGGTGATGCAGTAGAGAAA 277 ACCCAACTGAATGGAGCGGGTG 469
 1 A ATGCAGTAGAGAAAA Reverse Reverse sequence Reverse sequence primer name
 (5' -> 3') w/UT 207R_FtA_U TACCAGATGAACGAATAGC 278
ACGCACTTGACTTGTCTTCTACC 470 T2 C AGATGAACGAATAGCC

(122) All headings are for the convenience of the reader and should not be used to limit the meaning of the text that follows the heading, unless so specified.

(123) Unless defined otherwise, all technical and scientific terms herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although any methods and materials, similar or equivalent to those described herein, can be used in the practice or testing of the present invention, the preferred methods and materials are described herein. All publications, patents, and patent publications cited are incorporated by reference herein in their entirety for all purposes.

(124) The publications discussed herein are provided solely for their disclosure prior to the filing date of the present application. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such publication by virtue of prior invention.

(125) While the invention has been described in connection with specific embodiments thereof, it will be understood that it is capable of further modifications and this application is intended to cover any variations, uses, or adaptations of the invention following, in general, the principles of the invention and including such departures from the present disclosure as come within known or customary practice within the art to which the invention pertains and as may be applied to the essential features hereinbefore set forth and as follows in the scope of the appended claims.

Claims

1. A method of detecting *Bacillus anthracis* in a sample, comprising: generating one or more amplicons from the sample using at least one primer pair comprising a forward and reverse primer in at least one amplification reaction; sequencing the one or more amplicons using next-generation sequencing; and detecting at least one *B. anthracis*-specific amplicon in the sequenced amplicons, wherein each forward and reverse primer comprise a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 1 and SEQ ID NO: 2; SEQ ID NO: 3 and SEQ ID NO: 4; SEQ ID NO: 5 and SEQ ID NO: 6; SEQ ID NO: 7 and SEQ ID NO: 8; SEQ ID NO: 9 and SEQ ID NO: 10; SEQ ID NO: 11 and SEQ ID NO: 12; SEQ ID NO: 13 and SEQ ID NO: 14; SEQ ID NO: 15 and SEQ ID NO: 16; and SEQ ID NO: 17 and SEQ ID NO: 18; wherein the presence of the at least one *B. anthracis*-specific amplicon indicates the presence of *B. anthracis* in the sample, and the absence of the at least one *B. anthracis*-specific amplicon indicates the absence of *B. anthracis* from the sample.
2. The method of claim 1, further comprising confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 19 and SEQ ID NO: 20; SEQ ID NO: 21 and SEQ ID NO: 22; and SEQ ID NO: 23 and SEQ ID NO: 24; wherein detecting at least one *B. anthracis* Near Neighbor-specific amplicon in the sample confirms the absence of *B. anthracis*.
3. The method of claim 1, further comprising confirming the absence of *B. anthracis* by detecting at least one *B. anthracis* Near Neighbor-specific sequence variant (SV) or single nucleotide polymorphism (SNP) using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 25 and SEQ ID NO: 26; SEQ ID NO: 27 and SEQ ID NO: 28; SEQ ID NO: 49 and SEQ ID NO: 50; SEQ ID NO: 51 and SEQ ID NO: 52; SEQ ID NO: 53 and SEQ ID NO: 54; SEQ ID NO: 55 and SEQ ID NO: 56; SEQ ID NO: 57 and SEQ ID NO: 58; and SEQ ID NO: 59 and SEQ ID NO: 60; wherein detecting at least one *B. anthracis* Near Neighbor-specific SV in the sample confirms the absence of *B. anthracis*.
4. The method of claim 1, further comprising detecting a virulence locus or virulence plasmid in the sample by detecting a virulence-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group

consisting of: SEQ ID NO: 29 and SEQ ID NO: 30; SEQ ID NO: 31 and SEQ ID NO: 32; SEQ ID NO: 33 and SEQ ID NO: 34; and SEQ ID NO: 35 and SEQ ID NO: 36; wherein the presence of the virulence-specific amplicon indicates the presence of the virulence locus or virulence plasmid in the sample.

5. The method of claim 1, further comprising detecting at least one drug resistance single nucleotide polymorphism (SNP) from *B. anthracis* in the sample using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 37 and SEQ ID NO: 38; SEQ ID NO: 39 and SEQ ID NO: 40; SEQ ID NO: 41 and SEQ ID NO: 42; SEQ ID NO: 43 and SEQ ID NO: 44; SEQ ID NO: 45 and SEQ ID NO: 46; and SEQ ID NO: 47 and SEQ ID NO: 48.

6. The method of claim 1, further comprising detecting *Burkholderia pseudomallei* and/or *Burkholderia mallei* in the sample by detecting at least one *B. pseudomallei* or *B. mallei*-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 61 and SEQ ID NO: 62; SEQ ID NO: 63 and SEQ ID NO: 64; SEQ ID NO: 65 and SEQ ID NO: 66; SEQ ID NO: 67 and SEQ ID NO: 68; SEQ ID NO: 69 and SEQ ID NO: 70; SEQ ID NO: 71 and SEQ ID NO: 72; SEQ ID NO: 73 and SEQ ID NO: 74; SEQ ID NO: 75 and SEQ ID NO: 76; SEQ ID NO: 77 and SEQ ID NO: 78; SEQ ID NO: 79 and SEQ ID NO: 80; SEQ ID NO: 81 and SEQ ID NO: 82; SEQ ID NO: 83 and SEQ ID NO: 84; SEQ ID NO: 85 and SEQ ID NO: 86; SEQ ID NO: 87 and SEQ ID NO: 88; SEQ ID NO: 89 and SEQ ID NO: 90; SEQ ID NO: 91 and SEQ ID NO: 92; SEQ ID NO: 93 and SEQ ID NO: 94; SEQ ID NO: 95 and SEQ ID NO: 96; SEQ ID NO: 97 and SEQ ID NO: 98; SEQ ID NO: 99 and SEQ ID NO: 100; SEQ ID NO: 101 and SEQ ID NO: 102; SEQ ID NO: 103 and SEQ ID NO: 104; SEQ ID NO: 103 and SEQ ID NO: 104; SEQ ID NO: 105 and SEQ ID NO: 106; SEQ ID NO: 107 and SEQ ID NO: 108; SEQ ID NO: 117 and SEQ ID NO: 118; SEQ ID NO: 119 and SEQ ID NO: 120; SEQ ID NO: 121 and SEQ ID NO: 122; SEQ ID NO: 123 and SEQ ID NO: 124; and SEQ ID NO: 125 and SEQ ID NO: 126; wherein the presence of the at least one *B. pseudomallei* or *B. mallei*-specific amplicon indicates the presence of *B. pseudomallei* and/or *B. mallei* in the sample, and an absence of the at least one *B. pseudomallei* or *B. mallei*-specific amplicon indicates an absence of *B. pseudomallei* and *B. mallei* in the sample.

7. The method of claim 6, further comprising confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 177 and SEQ ID NO: 178; SEQ ID NO: 179 and SEQ ID NO: 180; SEQ ID NO: 181 and SEQ ID NO: 182; SEQ ID NO: 183 and SEQ ID NO: 184; SEQ ID NO: 185 and SEQ ID NO: 186; SEQ ID NO: 187 and SEQ ID NO: 188; SEQ ID NO: 189 and SEQ ID NO: 190; SEQ ID NO: 191 and SEQ ID NO: 192; SEQ ID NO: 193 and SEQ ID NO: 194; SEQ ID NO: 195 and SEQ ID NO: 196; SEQ ID NO: 197 and SEQ ID NO: 198; SEQ ID NO: 199 and SEQ ID NO: 200; SEQ ID NO: 201 and SEQ ID NO: 202; SEQ ID NO: 203 and SEQ ID NO: 204; SEQ ID NO: 205 and SEQ ID NO: 206; SEQ ID NO: 207 and SEQ ID NO: 208; SEQ ID NO: 207 and SEQ ID NO: 208; SEQ ID NO: 209 and SEQ ID NO: 210; SEQ ID NO: 211 and SEQ ID NO: 212; SEQ ID NO: 213 and SEQ ID NO: 214; SEQ ID NO: 215 and SEQ ID NO: 216; SEQ ID NO: 217 and SEQ ID NO: 218; SEQ ID NO: 219 and SEQ ID NO: 220; SEQ ID NO: 221 and SEQ ID NO: 222; SEQ ID NO: 223 and SEQ ID NO: 224; SEQ ID NO: 225 and SEQ ID NO: 226; SEQ ID NO: 227 and SEQ ID NO: 228; and SEQ ID NO: 229 and SEQ ID NO: 230; wherein detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific amplicon in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

8. The method of claim 6, further comprising confirming the absence of *B. pseudomallei* and *B. mallei* by detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 109 and SEQ ID NO: 110;

SEQ ID NO: 111 and SEQ ID NO: 112; SEQ ID NO: 113 and SEQ ID NO: 114; and SEQ ID NO: 115 and SEQ ID NO: 116; wherein detecting at least one *B. pseudomallei* or *B. mallei* Near Neighbor-specific SNP or SV in the sample confirms the absence of *B. pseudomallei* and *B. mallei*.

9. The method of claim 6, further comprising detecting at least one drug resistance SNP or SV from *Burkholderia* spp. in the sample using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 127 and SEQ ID NO: 128; SEQ ID NO: 129 and SEQ ID NO: 130; SEQ ID NO: 131 and SEQ ID NO: 132; SEQ ID NO: 133 and SEQ ID NO: 134; SEQ ID NO: 135 and SEQ ID NO: 136; SEQ ID NO: 137 and SEQ ID NO: 138; SEQ ID NO: 145 and SEQ ID NO: 146; SEQ ID NO: 147 and SEQ ID NO: 148; SEQ ID NO: 149 and SEQ ID NO: 150; SEQ ID NO: 151 and SEQ ID NO: 152; and SEQ ID NO: 153 and SEQ ID NO: 154.

10. The method of claim 1, further comprising detecting *Francisella tularensis* in the sample by detecting at least one *F. tularensis*-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 265 and SEQ ID NO: 266; SEQ ID NO: 267 and SEQ ID NO: 268; and SEQ ID NO: 269 and SEQ ID NO: 270; wherein the presence of the at least one *F. tularensis*-specific amplicon indicates that *F. tularensis* is present in the sample, and an absence of the at least one *F. tularensis*-specific amplicon indicates that *F. tularensis* is absent in the sample.

11. The method of claim 10, further comprising confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 285 and SEQ ID NO: 286; SEQ ID NO: 287 and SEQ ID NO: 288; SEQ ID NO: 289 and SEQ ID NO: 290; SEQ ID NO: 291 and SEQ ID NO: 292; and SEQ ID NO: 293 and SEQ ID NO: 294; wherein detecting at least one *F. tularensis* Near Neighbor-specific amplicon in the sample confirms the absence of *F. tularensis*.

12. The method of claim 10, further comprising confirming the absence of *F. tularensis* by detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 271 and SEQ ID NO: 272; SEQ ID NO: 273 and SEQ ID NO: 274; SEQ ID NO: 275 and SEQ ID NO: 276; SEQ ID NO: 277 and SEQ ID NO: 278; SEQ ID NO: 279 and SEQ ID NO: 280; SEQ ID NO: 281 and SEQ ID NO: 282; and SEQ ID NO: 283 and SEQ ID NO: 284; wherein detecting at least one *F. tularensis* Near Neighbor-specific SNP or SV in the sample confirms the absence of *F. tularensis*.

13. The method of claim 1, further comprising detecting *Yersinia pestis* in the sample by detecting at least one *Y. pestis*-specific amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 231 and SEQ ID NO: 232; SEQ ID NO: 233 and SEQ ID NO: 234; SEQ ID NO: 235 and SEQ ID NO: 236; and SEQ ID NO: 237 and SEQ ID NO: 238; wherein the presence of at least one *Y. pestis*-specific amplicon indicates the presence of *Y. pestis* in the sample, and an absence of at least one *Y. pestis*-specific amplicon indicates an absence of *Y. pestis* in the sample.

14. The method of claim 13, further comprising confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 249 and SEQ ID NO: 250; SEQ ID NO: 251 and SEQ ID NO: 252; SEQ ID NO: 253 and SEQ ID NO: 254; SEQ ID NO: 255 and SEQ ID NO: 256; SEQ ID NO: 257 and SEQ ID NO: 258; SEQ ID NO: 259 and SEQ ID NO: 260; and SEQ ID NO: 261 and SEQ ID NO: 262; wherein detecting at least one *Y. pestis* Near Neighbor-specific SNP or SV confirms the absence of *Y. pestis*.

15. The method of claim 13, further comprising confirming the absence of *Y. pestis* by detecting at least one *Y. pestis* Near Neighbor-specific amplicon using at least one primer pair comprising a

forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 263 and SEQ ID NO: 264; wherein detecting at least one *Y. pestis* Near Neighbor-specific amplicon confirms the absence of *Y. pestis*.

16. The method of claim 13, further comprising characterizing and/or subtyping *Y. pestis* in the sample by detecting at least one amplicon using at least one primer pair comprising a forward and reverse primer comprising a forward and reverse sequence selected from the group consisting of: SEQ ID NO: 239 and SEQ ID NO: 240; SEQ ID NO: 241 and SEQ ID NO: 242; SEQ ID NO: 243 and SEQ ID NO: 244; SEQ ID NO: 245 and SEQ ID NO: 246; and SEQ ID NO: 247 and SEQ ID NO: 248.

17. The method of claim 1, wherein the one or more amplicons are generated with at least one multiplex amplification reaction; and each primer in the at least one primer pair comprises a universal tail sequence.

18. The method of claim 17, wherein the at least one amplicon is present when a locus read count of the at least one amplicon is at least 10 sequence reads covering at least 75% of a corresponding amplicon reference sequence, and the universal tail sequence for each forward and reverse primer comprises SEQ ID NO: 301 and SEQ ID NO: 303.

19. The method of claim 1, wherein the sample is a biological sample obtained from a subject and further comprising administering an effective amount of at least one antibiotic to the subject, wherein the at least one antibiotic is selected from the group consisting of a fluoroquinolone, an aminoglycoside, a glycopeptide, a lincosamide, a macrolide/ketolide, a cephalosporin, a monobactam, a nitroimidazole, a penicillin, a streptogramin, a tetracycline, and a physiologically acceptable salt, prodrug, or combination thereof.

20. The method of 19, wherein the at least one antibiotic is not a fluoroquinolone if a *gyrA* drug resistance SNP is detected; and/or the at least one antibiotic is not a fluoroquinolone if a *parC* drug resistance SNP is detected; and/or the at least one antibiotic is not a fluoroquinolone or an aminocoumarin if a *gyrB* drug resistance SNP is detected; and/or the at least one antibiotic is not a rifamycin if a *rpoB* drug resistance SNP is detected; and/or the at least one antibiotic is not a β -lactam if a *penA* drug resistance SNP is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *folM* drug resistance SV is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeT* drug resistance SV is detected; and/or the at least one antibiotic is not a trimethoprim and sulfamethoxazole combination, co-trimoxazole, if a *bpeS* drug resistance SV is detected.
